

pid_multivalued_iv_vertex_calculus

Specialization: v1

TL;DR. In the finite response-type IV model with binary treatment, ordered n -point outcomes, l instrument arms, and no monotonicity, the gauge-fixed upper-dual section removes the $(l-1)$ -dimensional lineality and indexes the sharp ATE endpoint by M -distinct vertices. Exact activity signatures are recognizable in polynomial time, and the published arm-indexed family is strictly incomplete for every $n \geq 2$ and $l \geq 3$. The strongest explicit family certificate is now uniform in the number of arms when $n = 2$:

$$|\mathcal{V}_{2,l}| \geq 6 \binom{l}{2} + 9 \binom{l}{3} = \frac{3}{2} l^2 (l-1), \quad |\mathcal{V}_{2,l} \setminus \mathcal{F}_{2,l}^{\text{BBAK}}| \geq \frac{l(l-1)(3l-2)}{2}.$$

Its $l = 3$ specialization is exactly the certified 27-vertex total lower bound and 21-vertex comparator deficit. The sharp endpoints remain finite minima and maxima of observable linear functionals, but unconditional deterministic polynomial-delay enumeration for arbitrary (n, l) remains an explicit open problem.

1 Project justification

Gap. For multi-valued instruments, the existing analytical comparator constructions and projection treatments do not provide a complete list of M -distinct dual vertices, while generic inequality-based vertex enumeration expands a response-type system with $2^l n^2$ inequalities. The gap is both structural and computational: one must quotient the dual lineality, recognize exponentially encoded activity sets succinctly, quantify what published families omit, and still distinguish these advances from the unresolved inverse-child enumeration needed for deterministic polynomial delay.

Niche. The note develops exact vertex calculus on the gauge-fixed section $\mathcal{H}_{n,l}^0$. It combines a polynomial-time exact-signature recognizer, an explicit decreasing objective parent map, compressed tangent-ray verification, universal comparator-deficit certificates, and support-preserving zero-arm lifts. The new $n = 2$ result exploits disjoint exact essential-arm supports to place six certified vertices on every unordered arm pair and nine further certified vertices on every unordered arm triple.

Fill. The contribution supplies actionable finite linear-function representations of the sharp ATE endpoints indexed by $\mathcal{V}_{n,l}$, proves that the literal Boushehrian–Badri–Akbari–Kiyavash family is strictly incomplete for all $n \geq 2, l \geq 3$, retains the universal additive deficit theorem, and strengthens the concrete certificates to

$$|\mathcal{V}_{2,l}| \geq \frac{3}{2} l^2 (l-1), \quad |\mathcal{V}_{2,l} \setminus \mathcal{F}_{2,l}^{\text{BBAK}}| \geq \frac{l(l-1)(3l-2)}{2}.$$

It also isolates precisely what remains: converting the polynomial-time parent and local verification machinery into an unconditional polynomial-delay child iterator with polynomial-time exhaustion.

2 Related work

Boushehrian–Badri–Akbari–Kiyavash (arXiv:2604.12802) give an arm-indexed family of valid discrete-IV bounds and explicitly defer the multi-arm completeness problem. Song–Guo–Chan–Richardson (arXiv:2405.09510) likewise leave the relevant projection enumeration beyond the binary-instrument case unresolved. The present result does not merely compare formulas: it works in the quotient-correct gauge section, proves the literal arm-indexed family is a strict subset of the vertex set, and quantifies a universal deficit. For $n = 2$, its zero-arm lifts and arm permutations yield a new all- l lower bound with $9 \binom{l}{3}$ certified three-essential-arm vertices in addition to $6 \binom{l}{2}$ pair-essential vertices. The binary-instrument count is retained only as a consistency input, not promoted to an unproved multi-arm counting formula. The algorithmic discussion also respects the general hardness of vertex enumeration from explicit inequalities: polynomial-time recognition and parent evaluation do not by themselves establish polynomial-delay enumeration.

3 Setup

Target (estimand / structural parameter): $\theta = \mathbb{E}[Y(1) - Y(0)]$. Identifying functional / recovery map: $[L(p), U(p)] = [-U(S(p)), \min_{\lambda \in \mathcal{V}_{n,l}} \lambda^\top p]$ for $p \in \mathcal{P}_{n,l}^{\text{IV}} \subseteq \mathcal{C}_{n,l}$.

- n — integer; $n \geq 2$; support size

Definition. The number of ordered outcome values.

- l — integer; $l \geq 2$; support size

Definition. The number of instrument arms.

- $\mathcal{Y}_n$ — finite set; outcome support

Definition. $\mathcal{Y}_n = \{0, \dots, n-1\}$

- $\mathcal{D}$ — finite set; treatment support

Definition. $\mathcal{D} = \{0, 1\}$

- $\mathcal{Z}_l$ — finite set; instrument support

Definition. $\mathcal{Z}_l = \{1, \dots, l\}$

- Y — random variable; $Y \in \mathcal{Y}_n$; observed outcome

Definition. The observed outcome.

- D — random variable; $D \in \mathcal{D}$; observed treatment

Definition. The observed treatment.

- Z — random variable; $Z \in \mathcal{Z}_l$; observed instrument

Definition. The observed instrument.

- $Y(d)$ — potential outcome; $Y(d) \in \mathcal{Y}_n$ for $d \in \mathcal{D}$; causal outcome

Definition. The potential outcome under treatment d .

- δ — function; $\delta: \mathcal{Z}_l \rightarrow \mathcal{D}$; response component

Definition. A treatment-response word.

- η — function; $\eta: \mathcal{D} \rightarrow \mathcal{Y}_n$; response component

Definition. An outcome-response pair.

- τ — ordered pair; canonical response type

Definition. $\tau = (\delta, \eta)$.

- $\mathcal{T}_{n,l}$ — finite set; response-type space

Definition. $\mathcal{T}_{n,l} = \mathcal{D}^{\mathcal{Z}_l} \times \mathcal{Y}_n^{\mathcal{D}}$.

- $\mathcal{I}_{n,l}$ — finite set; observed-cell index set

Definition. $\mathcal{I}_{n,l} = \mathcal{Y}_n \times \mathcal{D} \times \mathcal{Z}_l$.

- p — vector; $p \in \mathbb{R}^{\mathcal{I}_{n,l}}$; observed conditional law

Definition. $p_{y,d,z} = \Pr(Y = y, D = d \mid Z = z)$.

- $\mathcal{C}_{n,l}$ — polytope; observed-law domain

Definition. The product conditional-law polytope with one normalized conditional cell law for each instrument arm.

- q — vector; $q \in \Delta(\mathcal{T}_{n,l})$; latent law

Definition. One probability law of response types, with $q \geq 0$ and $\mathbf{1}^\top q = 1$.

- $A_{\{n,l\}}$ — zero-one matrix; $A_{n,l} \in \{0,1\}^{\mathcal{I}_{n,l} \times \mathcal{T}_{n,l}}$; observed-to-latent map

Definition. The response-type incidence matrix.

- $c_{\{n,l\}}$ — integer vector; $c_{n,l} \in \mathbb{Z}^{\mathcal{T}_{n,l}}$; primal objective

Definition. The ATE loading on response types.

- $\mathcal{Q}(p)$ — polytope; primal feasible set

Definition. The compatible latent response-type laws.

- $\mathcal{P}^{\{\mathrm{IV}\}}_{\{n,l\}}$ — class; IV-law class

Definition. The class of conditional cell laws admitting a compatible response-type law.

- λ — vector; $\lambda \in \mathbb{R}^{\mathcal{I}_{n,l}}$; dual variable

Definition. An upper-dual coefficient vector.

- $\mathcal{H}_{\{n,l\}}$ — polyhedron; dual object

Definition. The upper-ATE dual feasible polyhedron.

- $K_{\{n,l\}}$ — linear subspace; dual lineality

Definition. $K_{n,l} = \ker(A_{n,l}^\top)$.

- $\mathcal{G}_{\{n,l\}}$ — linear subspace; transversal

Definition. The gauge section fixing $\lambda_{(0,0,z)} = 0$ for $z = 2, \dots, l$.

- $\mathcal{H}^0_{\{n,l\}}$ — polyhedron; pointed section

Definition. $\mathcal{H}^0_{n,l} = \mathcal{H}_{n,l} \cap \mathcal{G}_{n,l}$.

- $\sim$ — equivalence relation; M-equivalence

Definition. $\lambda \sim \lambda' \iff A_{n,l}^\top \lambda = A_{n,l}^\top \lambda'$.

- Γ — map; $\Gamma : \mathcal{H}_{n,l} \rightarrow \mathcal{H}^0_{n,l}$; quotient section

Definition. The canonical gauge representative map.

- $\mathcal{V}_{\{n,l\}}$ — finite set; M-distinct vertex index

Definition. $\mathcal{V}_{n,l} = \text{vert}(\mathcal{H}^0_{n,l})$.

- Sig — set-valued map; $\text{Sig} : \mathcal{H}_{n,l} \rightrightarrows \mathcal{T}_{n,l}$; exact activity signature

Definition. The set of tight response-type inequalities.

- B — finite list; $B \subseteq \mathcal{T}_{n,l}$; recognition input

Definition. An explicitly encoded proposed activity signature.

- r — integer; gauge-fixed ambient dimension

Definition. $r = 2nl - l + 1$.

- $\text{VERTREC}_{\{n,l\}}$ — decision problem; enumeration primitive

Definition. Exact recognition of whether B is the complete signature of a gauge-fixed vertex.

- $\text{E}_{\{n,l\}}$ — algorithm; open algorithmic target

Definition. A prospective rational-output enumerator of $\mathcal{V}_{n,l}$.

- $w(a, s)$ — vector; $w(a, s) \in \mathbb{R}^{\mathcal{I}_{n,l}}$; published comparator

Definition. The arm-indexed comparator vector of Boushehrian et al. in their lower-dual sign convention.

- $\bar{w}(a, s)$ — vector; $\bar{w}(a, s) \in \mathbb{R}^{\mathcal{I}_{n,l}}$; upper-dual comparator representative

Definition. The treatment-swapped negative of the published lower-dual vector.

- $\mathcal{F}_{\{n,l\}}$ — finite set; comparator family

Definition. The gauge-fixed upper-dual representatives of the published arm-indexed M-distinct family.

- I — map; $I : \mathbb{R}^{\mathcal{I}_{2,3}} \rightarrow \mathbb{R}^3$; gauge-invariant exclusion certificate

Definition. The treatment-one within-arm contrast vector.

- e_a — vector; $e_a \in \mathbb{R}^3$ for $a \in \{1, 2, 3\}$; comparator invariant value

Definition. The a -th standard basis vector.

- $\lambda^\dagger$ — rational vector; $\lambda^\dagger \in \mathbb{Q}^{\mathcal{I}_{2,3}}$; certified omitted vertex

Definition. The displayed gauge-fixed three-arm witness.

- λ' — rational vector; $\lambda' \in \mathbb{Q}^{\mathcal{I}_{2,3}}$; reverse-search diagnostic

Definition. The neighboring vertex in the worked basis exchange.

- $B^\dagger$ — ordered list; $B^\dagger \subseteq \mathcal{T}_{2,3}$; vertex certificate

Definition. The specified ten-equation active basis of $\lambda^\dagger$.

- B' — ordered list; $B' \subseteq \mathcal{T}_{2,3}$; pivot target

Definition. The canonical basis at the worked neighboring vertex.

- θ — real number; causal estimand

Definition. $\theta = \mathbb{E}[Y(1) - Y(0)]$.

- $U(p)$ — real number; identified-set endpoint

Definition. The sharp upper ATE endpoint.

- $L(p)$ — real number; identified-set endpoint

Definition. The sharp lower ATE endpoint.

- S — linear map; $S : \mathbb{R}^{\mathcal{I}_{n,l}} \rightarrow \mathbb{R}^{\mathcal{I}_{n,l}}$; endpoint symmetry

Definition. Treatment-label swapping of observed cell laws.

4 Assumptions

Assumption 1 (ass:finite-response-type-iv). *There exists $q \in \Delta(\mathcal{T}_{n,l})$ such that $A_{n,l}q = p$.*

Standard (finite response-type IV representation under exclusion, random assignment, and consistency, [2]).

5 Definitions

Definition 1 (def:iv-matrix). Defined object. $A_{n,l}$

Construction. $(A_{n,l})_{(y,d,z),(\delta,\eta)} = \mathbf{1}\{\delta(z) = d, \eta(d) = y\}$ for every $(y,d,z) \in \mathcal{I}_{n,l}$ and $(\delta,\eta) \in \mathcal{T}_{n,l}$.

Inputs. $\backslash\textit{mathcal I}_{\{n,l\}}; \backslash\textit{mathcal T}_{\{n,l\}}; \backslash\textit{delta}; \backslash\textit{eta}$.

Definition 2 (def:ate-loading). Defined object. $c_{n,l}$

Construction. $(c_{n,l})_{(\delta,\eta)} = \eta(1) - \eta(0)$ for every $\tau = (\delta,\eta) \in \mathcal{T}_{n,l}$.

Inputs. $\backslash\textit{mathcal T}_{\{n,l\}}; \backslash\textit{eta}$.

Definition 3 (def:conditional-law-polytope). Defined object. $\mathcal{C}_{n,l}$

Construction.

$$\mathcal{C}_{n,l} = \{p \in \mathbb{R}_+^{\mathcal{I}_{n,l}} : \sum_{y \in \mathcal{Y}_n} \sum_{d \in \mathcal{D}} p_{y,d,z} = 1 \text{ for every } z \in \mathcal{Z}_l\}$$

. Consequently $\mathbf{1}^\top p = l$.

Inputs. $p; \backslash\textit{mathcal I}_{\{n,l\}}; \backslash\textit{mathcal Y}_n; \backslash\textit{mathcal D}; \backslash\textit{mathcal Z}_l$.

Definition 4 (def:iv-law-class). Defined object. $\mathcal{P}_{n,l}^{\text{IV}}$

Construction.

$$\mathcal{P}_{n,l}^{\text{IV}} = \{p \in \mathcal{C}_{n,l} : \exists q \in \Delta(\mathcal{T}_{n,l}) \text{ such that } A_{n,l}q = p\}$$

Member properties. *ass:finite-response-type-iv*.

Definition 5 (def:primal-feasible-set). Defined object. $\mathcal{Q}(p)$

Construction. $\mathcal{Q}(p) = \{q \in \mathbb{R}_+^{\mathcal{T}_{n,l}} : A_{n,l}q = p\}$. For $p \in \mathcal{C}_{n,l}$, every $q \in \mathcal{Q}(p)$ has $\mathbf{1}^\top q = 1$, since each column of $A_{n,l}$ contributes once in every arm; the normalization is therefore redundant in this displayed primal system.

Inputs. $p; \backslash\textit{mathcal C}_{\{n,l\}}; \backslash\textit{A}_{\{n,l\}}; \backslash\textit{mathcal T}_{\{n,l\}}$.

Definition 6 (def:upper-dual). Defined object. $\mathcal{H}_{n,l}$

Construction. $\mathcal{H}_{n,l} = \{\lambda \in \mathbb{R}^{\mathcal{I}_{n,l}} : A_{n,l}^\top \lambda \geq c_{n,l}\}$.

Inputs. $\backslash\textit{A}_{\{n,l\}}; \backslash\textit{c}_{\{n,l\}}; \backslash\textit{mathcal I}_{\{n,l\}}$.

Definition 7 (def:gauge-map). Defined object. Γ

Construction. For $\lambda \in \mathcal{H}_{n,l}$, let $v_z = -\lambda_{(0,0,z)}$ for $z = 2, \dots, l$, let $v_1 = \sum_{z=2}^l \lambda_{(0,0,z)}$, set $v_{(y,d,z)} = v_z$, and define $\Gamma(\lambda) = \lambda + v$. Thus $\mathcal{G}_{n,l} = \{\lambda : \lambda_{(0,0,z)} = 0 \text{ for } z = 2, \dots, l\}$ and $\mathcal{H}_{n,l}^0 = \mathcal{H}_{n,l} \cap \mathcal{G}_{n,l}$.

Inputs. $\backslash\textit{lambda}; \backslash\textit{mathcal H}_{\{n,l\}}; \backslash\textit{mathcal I}_{\{n,l\}}; \backslash\textit{l}$.

Definition 8 (def:signature). Defined object. $\text{Sig}(\lambda)$

Construction.

$$\text{Sig}(\lambda) = \{\tau \in \mathcal{T}_{n,l} : (A_{n,l}^\top \lambda)_\tau = (c_{n,l})_\tau\}$$

Inputs. $\backslash\textit{lambda}; \backslash\textit{A}_{\{n,l\}}; \backslash\textit{c}_{\{n,l\}}; \backslash\textit{mathcal T}_{\{n,l\}}$.

Definition 9 (def:recognition-language). Defined object. $\text{VERTREC}_{n,l}$

Construction.

$$\text{VERTREC}_{n,l}(B) = 1 \iff \exists \lambda \in \mathcal{V}_{n,l} \text{ with } \text{Sig}(\lambda) = B$$

Inputs. $B; \backslash\textit{mathcal V}_{\{n,l\}}; \backslash\textit{operatorname{Sig}}$.

Definition 10 (def:bbak-family). Defined object. $\mathcal{F}_{n,l}^{\text{BBAK}}$

Construction. For $a \in \mathcal{Z}_l$ and nonconstant $s = (s_0, \dots, s_{n-1}) \in (\mathcal{Z}_l \setminus \{a\})^n$, define $w(a, s)_{y,1,j} = 0$ for $j \neq a$, $w(a, s)_{y,1,a} = y - (n-1)$, $w(a, s)_{y,0,j} = 0$ for $j \notin \{a, s_y\}$, $w(a, s)_{y,0,s_y} = n-1-y$, and $w(a, s)_{y,0,a} = -(n-1)$. The paper states this family in the lower-dual convention. In the upper-dual convention define

$\mathcal{F}_{n,l}^{\text{BBAK}} = \{\Gamma(\bar{w}(a, s)) : a \in \mathcal{Z}_l, s \in (\mathcal{Z}_l \setminus \{a\})^n \text{ nonconstant}, \bar{w}(a, s)_{y,d,z} = -w(a, s)_{y,1-d,z}\}$
. The indexing list has $l((l-1)^n - (l-1))$ entries, but different indices can define the same vector because s_{n-1} multiplies a zero coefficient. As a set of distinct literal coefficient vectors, $|\mathcal{F}_{n,2}^{\text{BBAK}}| = 0$ and $|\mathcal{F}_{n,l}^{\text{BBAK}}| = l(l-1)^{n-1}$ for $l \geq 3$.

Inputs. $\backslash \text{mathcal } Z_l; n; l; w(a, s); \backslash \text{Gamma}$.

Definition 11 (def:invariant). Defined object. I

Construction. For $\lambda \in \mathbb{R}^{\mathcal{I}_{2,3}}$, define

$$I(\lambda) = (\lambda_{(1,1,1)} - \lambda_{(0,1,1)}, \lambda_{(1,1,2)} - \lambda_{(0,1,2)}, \lambda_{(1,1,3)} - \lambda_{(0,1,3)})$$

Inputs. $\backslash \text{lambda}; \backslash \text{mathcal } I_ \{n, l\}$.

Definition 12 (def:three-arm-witness). Defined object. $\lambda^\dagger$

Construction. For $(n, l) = (2, 3)$, set the $z = 1, z = 2, z = 3$ blocks, in order $((0, 0), (1, 0), (0, 1), (1, 1))$, to $(2, 1, 2, 1)$, $(0, -2, -2, 0)$, and $(0, 1, 0, 1)$, respectively.

Inputs. $\backslash \text{mathcal } I_ \{n, l\}$.

Definition 13 (def:worked-transition). Defined object. $(B^\dagger, B', \lambda')$

Construction. Order types lexicographically by the treatment word and then by $(\eta(0), \eta(1))$. Let

$B^\dagger = \{((000), (11)), ((001), (10)), ((001), (11)), ((010), (00)), ((011), (00)), ((011), (10)), ((100), (01)), ((100), (11)), ((110), (00)), ((110), (01))\}$
. Replace the leaving equation $((001), (10))$ by the entering equation $((010), (10))$, retain the other nine equations of $B^\dagger$, and append the gauges $\lambda_{(0,0,2)} = \lambda_{(0,0,3)} = 0$. The ten type equations have rank 10; with the two gauges the system has rank 12 and solution with blocks $(2, 1, 2, 1)$, $(0, -1, -2, 0)$, $(0, 0, 0, 0)$, denoted λ' . The armwise separable slack minimum is nonnegative, and its equality words give exact signature

$\{((000), (11)), ((001), (11)), ((010), (00)), ((010), (10)), ((011), (00)), ((011), (10)), ((100), (01)), ((100), (11)), ((101), (01)), ((101), (11)), ((110), (00)), ((110), (01)), ((111), (00)), ((111), (01))\}$
. The lexicographically least rank basis in this signature is

$B' = \{((000), (11)), ((001), (11)), ((010), (00)), ((010), (10)), ((011), (00)), ((011), (10)), ((100), (01)), ((100), (11)), ((110), (00)), ((110), (01))\}$
. Since $B^\dagger \prec B'$, the proposed decreasing-basis orientation directs this edge from B' to $B^\dagger$.

Inputs. $\backslash \text{lambda}^\dagger; \backslash \text{lambda}'; \backslash \text{operatorname}\{Sig\}$.

Definition 14 (def:reverse-search-handle). Defined object. $E_{n,l}$ handle

Construction. Encode a node by the lexicographically least ordered set of $r = 2nl - l + 1$ independent tight type equations on $\mathcal{G}_{n,l}$; equivalently, retain those r equations together with the $l-1$ fixed gauge equations for an ambient rank- $2nl$ certificate. Encode $\tau = (\delta, \eta)$ by the l -bit word $(\delta(1), \dots, \delta(l))$ and the two labels $(\eta(0), \eta(1))$. Reconstruct the rational vertex by exact linear algebra. For one type-equation exchange, parameterize the resulting affine line; for each $(\eta(0), \eta(1))$, minimize slack over δ armwise as $\sum_{z=1}^l \min_{d \in \mathcal{D}} \lambda_{(\eta(d), d, z)} - (\eta(1) - \eta(0))$, enumerate its minimizing treatment words at equality, and thereby test feasibility, list the complete signature, and reject rank failure or unlisted tight types without expanding $2^l n^2$ types. A candidate root is $\Gamma(\bar{w}(1, (2, \dots, 2, 3)))$. The required parent map chooses, among recognized one-exchange neighbors whose canonical basis is lexicographically smaller, the lexicographically first; its proof obligations are uniqueness, acyclicity, reachability to the root, and polynomially many local tests. Inputs are $1^n, 1^l$; delay counts bit operations, basis and type encodings, and rational output. The integer zero-one equation matrix, ATE right sides bounded by $n-1$, and determinant bounds control numerator and denominator lengths.

Inputs. $r; \backslash \text{mathcal } H^0_ \{n, l\}; \backslash \text{operatorname}\{Sig\}; w(a, s); \backslash \text{Gamma}$.

Definition 15 (def:sharp-endpoints). Defined object. $U(p), L(p)$

Construction.

$$U(p) = \max_{q \in \mathcal{Q}(p)} c_{n,l}^\top q = \min_{\lambda \in \mathcal{V}_{n,l}} \lambda^\top p$$

, and $L(p) = -U(S(p))$ where $S(p)_{y,d,z} = p_{y,1-d,z}$.

Inputs. p ; $\mathcal{Q}(p)$; $c_{n,l}$; $\mathcal{V}_{n,l}$.

6 Main results

Lemma 1 (lem:gauge-transversal). $K_{n,l} = \{v : v_{(y,d,z)} = a_z \text{ for all } (y,d,z) \in \mathcal{I}_{n,l}, \sum_{z=1}^l a_z = 0\}$, so $\dim K_{n,l} = l - 1$. For every $p \in \mathcal{C}_{n,l}$ and $v \in K_{n,l}$, $v^\top p = \sum_z a_z = 0$. The map Γ is constant on each $\sim$ -class and that class meets $\mathcal{G}_{n,l}$ once. Hence $\mathcal{H}_{n,l}^0$ is pointed and its vertices biject with the M -distinct vertices of $\mathcal{H}_{n,l}$.

Proof. Let $v \in K_{n,l}$. By the definition of $A_{n,l}$, for every $\tau = (\delta, \eta) \in \mathcal{T}_{n,l}$,

$$0 = (A_{n,l}^\top v)_\tau = \sum_{z=1}^l v_{(\eta(\delta(z)), \delta(z), z)}.$$

Fix $z \in \mathcal{Z}_l$ and $y_0, y_1 \in \mathcal{Y}_n$, and choose $\eta(0) = y_0$ and $\eta(1) = y_1$. With $\delta(j) = 0$ for every j , the preceding equality is

$$\sum_{j=1}^l v_{(y_0, 0, j)} = 0.$$

With $\delta(z) = 1$ and $\delta(j) = 0$ for $j \neq z$, it is

$$v_{(y_1, 1, z)} + \sum_{j \neq z} v_{(y_0, 0, j)} = 0.$$

Subtracting gives $v_{(y_1, 1, z)} = v_{(y_0, 0, z)}$. Because y_0 and y_1 were arbitrary, all $2n$ coordinates in arm z have a common value a_z . The all-zero treatment word then gives $\sum_z a_z = 0$. Conversely, if $v_{(y,d,z)} = a_z$ and $\sum_z a_z = 0$, then

$$(A_{n,l}^\top v)_\tau = \sum_{z=1}^l a_z = 0$$

for every τ . This proves the asserted description of $K_{n,l}$. The map from $\{a \in \mathbb{R}^l : \sum_z a_z = 0\}$ to $K_{n,l}$ is a linear bijection, so $\dim K_{n,l} = l - 1$.

If $p \in \mathcal{C}_{n,l}$, armwise normalization gives

$$v^\top p = \sum_{z=1}^l a_z \sum_{y \in \mathcal{Y}_n} \sum_{d \in \mathcal{D}} p_{y,d,z} = \sum_{z=1}^l a_z = 0.$$

For $\lambda \in \mathcal{H}_{n,l}$, write $g(\lambda) = \Gamma(\lambda) - \lambda$. Its arm constants are

$$g_z(\lambda) = -\lambda_{(0,0,z)} \quad (z \geq 2), \quad g_1(\lambda) = \sum_{z=2}^l \lambda_{(0,0,z)},$$

so $g(\lambda) \in K_{n,l}$. Hence $A_{n,l}^\top \Gamma(\lambda) = A_{n,l}^\top \lambda$, and $\Gamma(\lambda) \in \mathcal{H}_{n,l}^0$. If $\lambda' = \lambda + k$ for an element $k \in K_{n,l}$ with arm constants a_z , then $g_z(\lambda') = g_z(\lambda) - a_z$ for $z \geq 2$, while

$$g_1(\lambda') = g_1(\lambda) + \sum_{z=2}^l a_z = g_1(\lambda) - a_1.$$

Thus $\Gamma(\lambda') = \Gamma(\lambda)$. Conversely, if $\lambda + k \in \mathcal{G}_{n,l}$ with $k \in K_{n,l}$, the gauge equations force $a_z = -\lambda_{(0,0,z)}$ for $z \geq 2$, and $\sum_z a_z = 0$ then uniquely fixes a_1 . Therefore $\lambda + k = \Gamma(\lambda)$, so each equivalence class meets $\mathcal{G}_{n,l}$ exactly once.

The lineality space of the section is

$$\text{lin}(\mathcal{H}_{n,l}^0) = K_{n,l} \cap \mathcal{G}_{n,l}.$$

An element of this intersection has $a_z = 0$ for $z \geq 2$, and then $a_1 = 0$ because $\sum_z a_z = 0$. Hence the section is pointed. Finally, Γ induces an affine bijection from the quotient polyhedron $\mathcal{H}_{n,l}/K_{n,l}$ onto $\mathcal{H}_{n,l}^0$. To see directly that it preserves extremality, note that a nontrivial midpoint decomposition in either space is carried by this affine bijection to a nontrivial midpoint decomposition in the other. The section vertices therefore biject with the M -distinct quotient vertices, as claimed. $\square$

Justification. The gauge removes the exact dual lineality and makes signatures, vertices, and observed-law functionals canonical. *Gap.* Without the transversal, the upper-dual polyhedron has no ordinary vertices because of its $l - 1$ -dimensional lineality. *Consumer.* The recognition algorithm and all exact rational vertex certificates operate in this pointed section.

Proposition 1 (prop:background-sharp-endpoints). *For every $p \in \mathcal{P}_{n,l}^{\text{IV}} \subseteq \mathcal{C}_{n,l}$, $\{c_{n,l}^\top q : q \in \mathcal{Q}(p)\} = [L(p), U(p)]$, with $U(p) = \min_{\lambda \in \mathcal{V}_{n,l}} \lambda^\top p$ and $L(p) = -U(S(p))$. The primal equality $A_{n,l}q = p$ already implies $\mathbf{1}^\top q = 1$ arm by arm, so this is the dual of the normalization-free displayed primal system.*

Proof. Fix $p \in \mathcal{P}_{n,l}^{\text{IV}}$. Then $\mathcal{Q}(p) \neq \emptyset$. For every $q \in \mathcal{Q}(p)$ and every fixed $z \in \mathcal{Z}_l$,

$$\begin{aligned} 1 &= \sum_{y,d} p_{y,d,z} = \sum_{y,d} (A_{n,l}q)_{y,d,z} \\ &= \sum_{\tau \in \mathcal{T}_{n,l}} q_\tau \sum_{y,d} \mathbf{1}\{\delta_\tau(z) = d, \eta_\tau(d) = y\} = \sum_{\tau \in \mathcal{T}_{n,l}} q_\tau. \end{aligned}$$

Thus $A_{n,l}q = p$ and $q \geq 0$ already imply $\mathbf{1}^\top q = 1$. Consequently $\mathcal{Q}(p)$ is a nonempty compact convex polytope.

The upper primal program and its dual are

$$\max\{c_{n,l}^\top q : A_{n,l}q = p, q \geq 0\}, \quad \min\{\lambda^\top p : A_{n,l}^\top \lambda \geq c_{n,l}\}.$$

The dual is feasible: the constant choice $\lambda_{(y,d,z)} = (n-1)/l$ satisfies

$$(A_{n,l}^\top \lambda)_\tau = n-1 \geq \eta(1) - \eta(0) = (c_{n,l})_\tau.$$

The cited finite response-type duality lemma therefore gives attainment and

$$\max_{q \in \mathcal{Q}(p)} c_{n,l}^\top q = \min_{\lambda \in \mathcal{H}_{n,l}} \lambda^\top p.$$

By the gauge-transversal lemma, $\Gamma(\lambda) - \lambda \in K_{n,l}$ and $\Gamma(\lambda)^\top p = \lambda^\top p$. Hence

$$\min_{\lambda \in \mathcal{H}_{n,l}} \lambda^\top p = \min_{\lambda \in \mathcal{H}_{n,l}^0} \lambda^\top p.$$

The minimizer set is a nonempty face of the pointed polyhedron $\mathcal{H}_{n,l}^0$. Choose a nonempty face of minimal dimension inside it. A positive-dimensional minimal face of a polyhedron is an affine space and hence has nonzero lineality;

that is impossible inside a pointed polyhedron. The chosen face is therefore zero-dimensional, so it is a vertex of $\mathcal{H}_{n,l}^0$. It follows that

$$\max_{q \in \mathcal{Q}(p)} c_{n,l}^\top q = \min_{\lambda \in \mathcal{V}_{n,l}} \lambda^\top p = U(p).$$

For the lower endpoint, define the response-type involution

$$R(\delta, \eta) = (\tilde{\delta}, \tilde{\eta}), \quad \tilde{\delta}(z) = 1 - \delta(z), \quad \tilde{\eta}(d) = \eta(1 - d).$$

Given $q \in \mathcal{Q}(p)$, set $\tilde{q}_{R(\tau)} = q_\tau$. Direct substitution gives

$$(A_{n,l} \tilde{q})_{y,d,z} = \sum_{\tau} q_{\tau} \mathbf{1}\{\delta_{\tau}(z) = 1 - d, \eta_{\tau}(1 - d) = y\} = p_{y,1-d,z} = S(p)_{y,d,z},$$

and

$$(c_{n,l})_{R(\tau)} = \tilde{\eta}(1) - \tilde{\eta}(0) = \eta(0) - \eta(1) = -(c_{n,l})_{\tau}.$$

Since R is a bijection, $q \mapsto \tilde{q}$ is a bijection from $\mathcal{Q}(p)$ to $\mathcal{Q}(S(p))$. Therefore

$$\min_{q \in \mathcal{Q}(p)} c_{n,l}^\top q = - \max_{\tilde{q} \in \mathcal{Q}(S(p))} c_{n,l}^\top \tilde{q} = -U(S(p)) = L(p).$$

Finally, the continuous linear image of the nonempty compact convex set $\mathcal{Q}(p)$ is the entire compact interval between its minimum and maximum. Thus

$$\{c_{n,l}^\top q : q \in \mathcal{Q}(p)\} = [L(p), U(p)].$$

□

Justification. This recalled LP-duality fact explains why a complete vertex catalogue gives explicit sharp endpoints. *Gap.* It is established background from Balke–Pearl finite response-type duality and the dual formulation used by Boushehrian et al.; it is not a claimed contribution. *Consumer.* It translates any successful enumeration into sharp all-arm ATE evaluation.

Open-ended Question 1 (oeq:polynomial-delay-enumeration). *For $n \geq 2$ and $l \geq 3$, let P be the explicit polynomial-time parent map with root λ^* furnished by the global-parent theorem. Can one construct a deterministic rational-output child generator which, on input $\mu \in \mathcal{V}_{n,l}$, lists exactly the adjacent vertices ν satisfying $P(\nu) = \mu$ with bit-operation delay polynomial in $n + l$, reports exhaustion in polynomial time after its last child, and can be coupled to a traversal that lists every element of $\mathcal{V}_{n,l}$ exactly once with global bit-operation delay polynomial in $n + l$, including polynomial-time termination after the final vertex? As one sufficient route, can the relevant finite extreme rays of the tangent cone at each μ , their exact finite line-search endpoints, and the parent-filtered subset be generated with these delay and exhaustion guarantees without expanding all $2^l n^2$ type inequalities or all active bases?*

Unconditionally, P is computable in polynomial bit time, has the known root λ^* , strictly decreases its rational objective, and reaches the root from every vertex. Given a rational candidate h , one can decide in polynomial bit time whether h is a vertex of $\mathcal{K}_\mu$, whether its tangent ray has a finite exact endpoint ν , and whether $P(\nu) = \mu$, without expanding the $2^l n^2$ type inequalities. Every simple vertex, characterized here by $|\text{Sig}(\mu)| = r$, therefore admits exact polynomial-total-time child generation with polynomial-time exhaustion. For arbitrary vertices, every normalized ray has a canonical encoding of length polynomial in $n + l$, and NP-oracle prefix search lists every child exactly once with polynomial delay and polynomial-time exhaustion. Coupling this iterator to the parent tree through the parity–Euler traversal lists every element of $\mathcal{V}_{n,l}$ exactly once with global polynomial delay and polynomial-time final termination. Consequently the full requested enumerator exists relative to NP, and it exists deterministically if $P = \text{NP}$; unconditional oracle removal at nonsimple vertices is the sole residual.

Justification. A complete deterministic child iterator would turn the finite sharp endpoint representation into an unconditional polynomial-delay analytical enumerator. *Gap.* The inverse-child generation and polynomial-time exhaustion guarantees remain open at degenerate vertices, despite the global parent map, compressed verification, simple-vertex algorithm, and NP-oracle upper bound. *Consumer.* Vertex-enumeration software and multi-arm IV sharp-bound routines need this residual resolved before the note can claim unconditional polynomial-delay listing of every coefficient vector.

Theorem 1 (thm:polynomial-time-exact-signature-recognition). *For every $n \geq 2$ and $l \geq 2$, $\text{VERTREC}_{n,l}(B)$ is decidable by a deterministic algorithm in bit-operation time polynomial in $n + l + \text{enc}(B)$, where $\text{enc}(B)$ is the explicit bit length of the list. The algorithm solves the equations indexed by B in the $r = 2nl - l + 1$ gauge coordinates, tests the n^2 separable minima $m_{ab}(\lambda) = \sum_{z=1}^l \min\{\lambda_{(a,0,z)}, \lambda_{(b,1,z)}\}$, and certifies signature completeness by comparing the listed minimizing words with the product cardinalities $2^{t_{ab}}$, where t_{ab} is the number of tied arms.*

Proof. Work in the $r = 2nl - l + 1$ free coordinates of $\mathcal{G}_{n,l}$, inserting the $l - 1$ fixed zero coordinates whenever a type row is evaluated. First validate every encoded type in B and replace the input list by its set of distinct entries, because $\text{Sig}(\lambda)$ is a set. For $\tau = (\delta, \eta) \in B$, let $\alpha_\tau \in \{0, 1\}^r$ be the restriction to the free coordinates of the corresponding row of $A_{n,l}^\top$. Exact elimination is applied to

$$\alpha_\tau^\top x = \eta(1) - \eta(0), \quad \tau \in B.$$

Reject if this system is inconsistent or if its row rank is less than r . If it is consistent and has rank r , it has a unique rational solution; insert the fixed gauge coordinates and call the resulting vector λ .

We first justify why rank r is exactly the required vertex rank. Define $\lambda^\circ \in \mathcal{G}_{n,l}$ by setting all $2n$ coordinates in arm 1 equal to n , setting $\lambda_{(0,0,z)}^\circ = 0$ for $z \geq 2$, and setting every other free coordinate equal to 1. Every type sum receives n from arm 1 and a nonnegative contribution from every other arm. Since

$$(c_{n,l})_{(\delta,\eta)} = \eta(1) - \eta(0) \leq n - 1,$$

all the defining inequalities of $\mathcal{H}_{n,l}^0$ are strict at λ° . Thus $\mathcal{H}_{n,l}^0$ is full-dimensional in the r -dimensional gauge space. At a feasible point of a full-dimensional polyhedron, the point is a vertex if and only if its active normals span the ambient space. Indeed, spanning makes the common affine solution set of the active equations a singleton. If the active normals do not span, choose a nonzero gauge-space direction orthogonal to all of them. Finiteness of the defining system and strict positivity of every inactive slack permit a sufficiently small displacement in both signs along this direction, expressing the point as the midpoint of two distinct feasible points. This proves the criterion directly.

It remains to test feasibility and the complete signature without expanding the $2^l n^2$ type inequalities. Fix $(a, b) \in \mathcal{Y}_n^2$, and consider types satisfying $\eta(0) = a$ and $\eta(1) = b$. By the definition of the incidence matrix,

$$(A_{n,l}^\top \lambda)_{(\delta,\eta)} = \sum_{z=1}^l \lambda_{(\eta(\delta(z)), \delta(z), z)}.$$

The choices $\delta(z) \in \{0, 1\}$ are independent across arms. Therefore the elementary separability identity gives

$$\min_{\delta \in \mathcal{D}^{\mathcal{Z}_l}} \{(A_{n,l}^\top \lambda)_{(\delta,\eta)} - (b - a)\} = \sum_{z=1}^l \min\{\lambda_{(a,0,z)}, \lambda_{(b,1,z)}\} - (b - a) = m_{ab}(\lambda) - (b - a).$$

Consequently,

$$\lambda \in \mathcal{H}_{n,l}^0 \iff m_{ab}(\lambda) \geq b - a \text{ for every } (a, b) \in \mathcal{Y}_n^2.$$

The algorithm computes these n^2 sums and rejects if one of the inequalities fails.

Suppose feasibility holds. For each outcome pair and arm, define

$$R_{ab,z} = \operatorname{argmin}_{d \in \{0,1\}} \lambda_{(\eta(d), d, z)}, \quad \eta(0) = a, \quad \eta(1) = b.$$

If $m_{ab}(\lambda) > b - a$, no type with outcome pair (a, b) is tight. If $m_{ab}(\lambda) = b - a$, subtracting the separable minimum expresses the slack of a treatment word as a sum of nonnegative armwise gaps. That sum is zero if and only if

$$\delta(z) \in R_{ab,z} \quad \text{for every } z \in \mathcal{Z}_l.$$

Thus the tight treatment words for (a, b) are exactly the Cartesian product $\prod_{z=1}^l R_{ab,z}$. Each factor has one element except when $\lambda_{(a,0,z)} = \lambda_{(b,1,z)}$, when it has two. Hence, with

$$t_{ab} = \left| \left\{ z : \lambda_{(a,0,z)} = \lambda_{(b,1,z)} \right\} \right|,$$

the product contains exactly $2^{t_{ab}}$ words.

The completeness test groups the distinct members of B by (a, b) . In the strict case it requires the group to be empty. In the equality case it checks that every listed treatment word belongs to $\prod_z R_{ab,z}$ and that the group has cardinality $2^{t_{ab}}$. Inclusion together with equality of finite cardinalities is equivalent to equality with the entire product. The test therefore accepts exactly when $B = \text{Sig}(\lambda)$. If it accepts, the equations indexed by B have rank r , so the feasible candidate is a vertex by the criterion above. Conversely, if B is the complete signature of a vertex, its active normals have rank r , its equations yield that vertex as their unique solution, every separable feasibility test passes, and every product test recovers precisely its tight types. This proves soundness and completeness.

Finally, all steps have polynomial bit complexity. There are $r = O(nl)$ unknowns and $|B|$ explicitly supplied zero-one rows, each constructible in $O(l)$ operations. Choose any nonsingular r -row subsystem. Expanding its determinant gives absolute value at most $r!$; after replacing one column by a right-hand side whose entries have absolute value at most $n - 1$, the same expansion gives absolute value at most $(n - 1)r!$. Cramer's rule therefore bounds every numerator and denominator of the candidate by $O(r \log r + \log n)$ bits. Fraction-free exact elimination, the $O(n^2 l)$ rational comparisons defining the minima, and the $O(l|B|)$ product-membership checks consequently use polynomially many bit operations on polynomial-length integers. The integer $2^{t_{ab}}$ has at most $l + 1$ bits and is compared only with the cardinality of an explicitly encoded group of B ; no minimizing product is enumerated. The total running time is polynomial in $n + l + \text{enc}(B)$, as claimed. $\square$

Justification. Armwise separability recognizes complete tight-type signatures without expanding all $2^l n^2$ inequalities. *Gap.* Recognition of a proposed signature does not generate all signatures or all inverse-parent children with polynomial delay. *Consumer.* Exact rational implementations obtain a polynomial-time certificate for candidate gauge-fixed vertices and their complete activity sets.

Theorem 2 (thm:universal-bbak-family-deficit). *Let $N_n = 5 \cdot 4^{n-1} - 2^{n+2} + 4$. For every $n \geq 2$ and $l \geq 3$, exactly $l(l - 1)$ vertices in $\mathcal{F}_{n,l}^{\text{BBAK}}$ have two essential arm blocks, and its remaining $l((l - 1)^{n-1} - (l - 1))$ vertices have at least three essential arm blocks. Consequently,*

$$|\mathcal{V}_{n,l}| \geq \binom{l}{2}(N_n - 2) + l((l - 1)^{n-1} - (l - 1)) = |\mathcal{F}_{n,l}^{\text{BBAK}}| + \frac{l(l - 1)}{2}(N_n - 4).$$

Moreover,

$$|\mathcal{V}_{n,l} \setminus \mathcal{F}_{n,l}^{\text{BBAK}}| \geq \binom{l}{2}(N_n - 2) - l(l - 1) = \frac{l(l - 1)}{2}(N_n - 4) > 0.$$

Thus the literal arm-indexed family is strictly incomplete for every $n \geq 2$ and $l \geq 3$.

Proof. Fix $n \geq 2$ and $l \geq 3$, and put $m = n - 1$ and $N_n = 5 \cdot 4^{n-1} - 2^{n+2} + 4$. Call an arm essential when its block of $2n$ coefficients is nonconstant. The essential-pair vertex lower-bound lemma shows both that essential-arm support is invariant under gauge fixing and that there is a set $\mathcal{W}_{n,l} \subseteq \mathcal{V}_{n,l}$ satisfying

$$|\mathcal{W}_{n,l}| \geq \binom{l}{2}(N_n - 2), \quad |\text{supp}_{\text{ess}}(\lambda)| = 2 \quad (\lambda \in \mathcal{W}_{n,l}). \quad (1)$$

We next partition the literal comparator family by the same invariant. The comparator vertex-and-count lemma proves that every distinct comparator is uniquely indexed by a distinguished arm a and an effective prefix

$$(s_0, \dots, s_{m-1}) \in (\mathcal{Z}_l \setminus \{a\})^m. \quad (2)$$

The within-arm contrast formulas established there are, before gauge fixing,

$$\bar{w}(a, s)_{(y,0,z)} = (m - y)\mathbf{1}\{z = a\}, \quad (3)$$

$$\bar{w}(a, s)_{(y,1,z)} = m\mathbf{1}\{z = a\} + (y - m)\mathbf{1}\{z = s_y\}. \quad (4)$$

Gauge fixing adds only a constant to an entire arm block, so it cannot change whether that block is constant. Equation (3) shows that arm a is essential because $m > 0$. If $z \neq a$ and $z = s_y$ for some $y < m$, then (4) has value $y - m \neq 0$ at the treatment-one coordinate indexed by y , whereas its treatment-one coordinate indexed by m is

zero; hence arm z is essential. Conversely, if $z \neq a$ does not occur among $s_0, \dots, s_{m-1}$, then (3) is zero throughout arm z , while (4) is also zero there for every $y < m$ and its coefficient at $y = m$ is zero regardless of s_m . Thus

$$\text{supp}_{\text{ess}}(\Gamma(\bar{w}(a, s))) = \{a\} \cup \{s_0, \dots, s_{m-1}\}. \quad (5)$$

Because every s_y in (2) differs from a , the support in (5) has cardinality two exactly when the effective prefix is constant. There are l choices of a and then $l - 1$ choices for the constant prefix. Hence exactly

$$l(l - 1) \quad (6)$$

comparator vertices have two essential arms. The comparator-count lemma gives $|\mathcal{F}_{n,l}^{\text{BBAK}}| = l(l - 1)^m$, so the remaining comparator subfamily, denoted $\mathcal{F}_{\geq 3}$, has cardinality

$$|\mathcal{F}_{\geq 3}| = l((l - 1)^{n-1} - (l - 1)), \quad (7)$$

and every one of its members has at least three essential arms.

The comparator lemma gives $\mathcal{F}_{\geq 3} \subseteq \mathcal{V}_{n,l}$. By (1), every member of $\mathcal{W}_{n,l}$ has exactly two essential arms, whereas every member of $\mathcal{F}_{\geq 3}$ has at least three. The two subsets are therefore disjoint. Taking the cardinality of their union inside $\mathcal{V}_{n,l}$ and using (1) and (7) yields the additive bound

$$|\mathcal{V}_{n,l}| \geq \binom{l}{2}(N_n - 2) + l((l - 1)^{n-1} - (l - 1)). \quad (8)$$

For the deficit itself, (6) implies

$$|\mathcal{W}_{n,l} \cap \mathcal{F}_{n,l}^{\text{BBAK}}| \leq l(l - 1).$$

Consequently,

$$\begin{aligned} |\mathcal{V}_{n,l} \setminus \mathcal{F}_{n,l}^{\text{BBAK}}| &\geq |\mathcal{W}_{n,l} \setminus \mathcal{F}_{n,l}^{\text{BBAK}}| \\ &\geq \binom{l}{2}(N_n - 2) - l(l - 1) \\ &= \frac{l(l - 1)}{2}(N_n - 4). \end{aligned} \quad (9)$$

Equivalently, since $|\mathcal{F}_{n,l}^{\text{BBAK}}| = l(l - 1)^{n-1}$, equation (8) is

$$|\mathcal{V}_{n,l}| \geq |\mathcal{F}_{n,l}^{\text{BBAK}}| + \frac{l(l - 1)}{2}(N_n - 4). \quad (10)$$

Finally, if $x = 2^{n-1} \geq 2$, then

$$N_n - 4 = 5 \cdot 4^{n-1} - 2^{n+2} = x(5x - 8) \geq 4.$$

Thus the right side of (9) is positive, proving strict incompleteness for every $n \geq 2$ and $l \geq 3$. The universal two-arm omitted-vertex lemma additionally supplies a concrete omitted vertex for every such pair; equation (9) strengthens that witness into the asserted cardinality deficit. $\square$

Justification. The universal theorem continues to prove strict comparator incompleteness for every $n \geq 2$ and $l \geq 3$, including regimes not covered by the sharper binary-outcome construction. *Gap.* Its additive deficit remains a certified lower bound and is not an exact formula for $|\mathcal{V}_{n,l}|$. *Consumer.* Software that restricts optimization to the literal comparator family is warned that this restriction is non-sharp throughout the full multi-valued-instrument regime.

Lemma 2 (lem:finite-response-type-duality). *For the finite response-type IV program $\max\{c_{n,l}^\top q : A_{n,l}q = p, q \geq 0\}$ and its dual $\min\{\lambda^\top p : A_{n,l}^\top \lambda \geq c_{n,l}\}$, whenever the primal feasible set is nonempty, the primal and dual optimal values are attained and equal.*

Theorem 3 (thm:decreasing-basis-root-obstruction). *The decreasing-canonical-basis parent rule in the reverse-search handle does not yield the claimed rooted enumeration, already for $(n, l) = (2, 3)$. Its proposed root $\rho = \Gamma(\bar{w}(1, (2, 3)))$ has canonical basis strictly larger than the canonical basis of another vertex $\xi \in \mathcal{V}_{2,3}$; hence no parent chain that strictly decreases canonical bases can lead from ξ to ρ . Thus the specified construction does not establish a polynomial-delay enumerator. This obstruction does not rule out a different polynomial-delay algorithm.*

Proof. Write $u_{y,z} = \lambda_{(y,0,z)}$ and $v_{y,z} = \lambda_{(y,1,z)}$, and order each arm block as $(u_{0,z}, u_{1,z}, v_{0,z}, v_{1,z})$. For $n = 2$, $a = 1$, and $s = (2, 3)$, direct substitution in the definition of $w(a, s)$, followed by treatment swapping and negation, gives the proposed root

$$\rho = \Gamma(\bar{w}(1, (2, 3))) = ((1, 0, 1, 1), (0, 0, -1, 0), (0, 0, 0, 0)).$$

Interchanging arms two and three in the indexing word gives the distinct gauge-fixed vector

$$\xi = \Gamma(\bar{w}(1, (3, 2))) = ((1, 0, 1, 1), (0, 0, 0, 0), (0, 0, -1, 0)).$$

Both vectors already have $u_{0,2} = u_{0,3} = 0$, so the displayed applications of Γ do not alter them.

Let $\eta = (a, b) \in \{0, 1\}^2$. By the definitions of $A_{2,3}$ and $c_{2,3}$, the slack of the type (δ, η) at $\alpha \in \{\rho, \xi\}$ is

$$s_{\delta,ab}^{\alpha} = \sum_{z=1}^3 \alpha_{(\eta(\delta(z)), \delta(z), z)} - (b - a).$$

Exact substitution yields all thirty-two slacks for each vector:

$\delta \backslash \eta$	00	01	10	11
000	1	0	1	0
001	1	0	1	0
010	0	0	0	0
011	0	0	0	0
100	1	0	2	1
101	1	0	2	1
110	0	0	1	1
111	0	0	1	1

$(\alpha = \rho),$

$\delta \backslash \eta$	00	01	10	11
000	1	0	1	0
001	0	0	0	0
010	1	0	1	0
011	0	0	0	0
100	1	0	2	1
101	0	0	1	1
110	1	0	2	1
111	0	0	1	1

$(\alpha = \xi).$

Every entry is nonnegative. Therefore $A_{2,3}^\top \rho \geq c_{2,3}$ and $A_{2,3}^\top \xi \geq c_{2,3}$; together with the gauge equalities, this proves $\rho, \xi \in \mathcal{H}_{2,3}^0$. The zero entries in the two tables are exactly their complete signatures.

The gauge section has dimension $r = 2 \cdot 2 \cdot 3 - 3 + 1 = 10$; the gauge-transversal lemma ensures that there is no residual lineality. Scan each complete signature in the prescribed lexicographic order and retain a type equation precisely when its normal increases the rank in the gauge coordinates. Writing δ/ab for $(\delta, (a, b))$, the greedy scans retain

$$B_\rho = (000/01, 000/11, 001/01, 001/11, 010/00, 010/01, \\ 010/10, 011/00, 100/01, 110/00),$$

and

$$B_\xi = (000/01, 000/11, 001/00, 001/01, 001/10, 010/01, \\ 010/11, 011/00, 100/01, 101/00).$$

For a fully exact rank certificate, use the reduced coordinate order

$$x = (u_{0,1}, u_{1,1}, v_{0,1}, v_{1,1}, u_{1,2}, v_{0,2}, v_{1,2}, u_{1,3}, v_{0,3}, v_{1,3}).$$

The row-normal matrices of the retained systems are

$$R_\rho = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \det(R_\rho) = -1,$$

and

$$R_\xi = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}, \quad \det(R_\xi) = 1.$$

Thus each feasible vector has ten linearly independent tight inequalities in the ten-dimensional gauge section. To verify extremality directly, suppose one of them, denoted α , were the midpoint of feasible section points α^- and α^+ . Every inequality tight at α has nonnegative slack at both endpoints and average slack zero, so it is tight at both endpoints. Hence $R_\alpha(\alpha^+ - \alpha^-) = 0$ in reduced coordinates. The displayed nonzero determinant implies $\alpha^+ = \alpha^-$. Consequently $\rho, \xi \in \mathcal{V}_{2,3}$.

It remains to justify that the two retained lists are the canonical bases, not merely full-rank bases. For any linearly ordered finite list of row vectors, the greedy scan retaining the first row that increases the span returns the lexicographically least full-rank sublist. Indeed, at the first position where a competing full-rank sublist differs from the greedy list, the competing list cannot contain an earlier row omitted by the greedy procedure, while omission of the greedy row forces its eventual replacement by a strictly later row. Applying this argument successively to the two ordered signatures proves that B_ρ and B_ξ are exactly the canonical bases specified by the reverse-search handle.

The first two entries of these bases agree. Their third entries are, respectively, 001/01 and 001/00, and the prescribed lexicographic type order has 001/00 $\prec$ 001/01. Therefore

$$B_\xi \prec B_\rho.$$

Every parent edge under the proposed rule strictly decreases the canonical basis. By induction, every vertex reached after one or more parent steps from ξ has canonical basis strictly smaller than B_ξ , and therefore cannot equal ρ ,

whose canonical basis is strictly larger. Thus no parent chain from ξ reaches the proposed root ρ . Root reachability fails already for $(n, l) = (2, 3)$, so the decreasing-canonical-basis rule in the reverse-search handle does not define the claimed rooted enumeration and does not establish a polynomial-delay enumerator. The argument concerns only this parent orientation and does not preclude a different polynomial-delay algorithm. $\square$

Justification. The explicit counterexample prevents the failed parent rule from being mistaken for a complete reverse-search proof. *Gap.* It refutes one orientation only and leaves alternative polynomial-delay algorithms unresolved. *Consumer.* Future enumeration work must use a different globally reachable parent certificate or traversal.

Lemma 3 (lem:rational-linear-programming-polynomial-time). *A finite rational linear program is exactly solvable by a deterministic algorithm in time polynomial in the numbers of variables and inequalities and in the binary encoding length of its coefficients.*

Theorem 4 (thm:explicit-global-parent-map). *For every $n \geq 2$ and $l \geq 3$, the vertex set $\mathcal{V}_{n,l}$ admits an explicit deterministic parent map with a known root. In the notation $u_{a,z} = \lambda_{(a,0,z)}$ and $v_{b,z} = \lambda_{(b,1,z)}$, the root λ^* is given by $u_{a,1}^* = n - 1$, $v_{b,1}^* = b$, $u_{a,2}^* = -a$, $v_{b,2}^* = 0$, and $u_{a,z}^* = v_{b,z}^* = 0$ for $z \geq 3$. There is a rational linear objective g , constructible from a polynomial-size set of active type rows at λ^* , for which λ^* is the unique minimizer over $\mathcal{H}_{n,l}^0$. For each $\mu \in \mathcal{V}_{n,l} \setminus \{\lambda^*\}$, the map computes in bit-operation time polynomial in $n + l$ an adjacent vertex $P(\mu)$ satisfying $g^\top P(\mu) < g^\top \mu$. Hence the directed parent graph is acyclic and every parent chain terminates at λ^* . This result does not assert polynomial-delay enumeration of the inverse child sets of P .*

Proof. Write $u_{a,z} = \lambda_{(a,0,z)}$ and $v_{b,z} = \lambda_{(b,1,z)}$. For a type with $\eta(0) = a$, $\eta(1) = b$, the definitions of $A_{n,l}$ and $c_{n,l}$ give

$$(A_{n,l}^\top \lambda)_{(\delta,\eta)} - (c_{n,l})_{(\delta,\eta)} = \sum_{z=1}^l (\mathbf{1}\{\delta(z) = 0\} u_{a,z} + \mathbf{1}\{\delta(z) = 1\} v_{b,z}) - (b - a).$$

Minimization over the treatment word is armwise, so

$$\lambda \in \mathcal{H}_{n,l}^0 \iff \lambda \in \mathcal{G}_{n,l} \text{ and } \sum_{z=1}^l \min\{u_{a,z}, v_{b,z}\} \geq b - a \text{ for all } (a, b) \in \mathcal{Y}_n^2. \quad (1)$$

Define λ^* by the formula in the statement. It obeys the gauge equations because $u_{0,z}^* = 0$ for every $z \geq 2$. For every (a, b) ,

$$\sum_{z=1}^l \min\{u_{a,z}^*, v_{b,z}^*\} = \min\{n - 1, b\} + \min\{-a, 0\} = b - a. \quad (2)$$

Thus $\lambda^* \in \mathcal{H}_{n,l}^0$, and every outcome-pair minimum in (1) is attained with equality.

We next give a polynomial-size rank certificate. For every (a, b) , let the default active type have $\eta = (a, b)$, $\delta(1) = 1$, $\delta(2) = 0$, and $\delta(z) = 0$ for $z \geq 3$. Include also the active type obtained by flipping only $\delta(z)$ for each $z \geq 3$. When $b = n - 1$, include the type obtained from the default by flipping only $\delta(1)$; when $a = 0$, include the type obtained by flipping only $\delta(2)$. Denote this pool by $\mathcal{R}^*$. Equation (2) shows that all of these types are active, and

$$|\mathcal{R}^*| \leq n^2(l - 1) + 2n. \quad (3)$$

Let $h \in \mathcal{G}_{n,l}$ be a gauge direction annihilated by every row in $\mathcal{R}^*$. Subtracting a later-arm flip equation from its default equation gives

$$h_{(b,1,z)} = h_{(a,0,z)} \quad \text{for all } a, b \in \mathcal{Y}_n \text{ and } z \geq 3.$$

For fixed $z \geq 3$, all coordinates in this display have a common value, and the gauge coordinate $h_{(0,0,z)} = 0$ makes that value zero. The default equations therefore reduce to

$$h_{(b,1,1)} + h_{(a,0,2)} = 0 \quad \text{for all } a, b.$$

Taking $a = 0$ and using $h_{(0,0,2)} = 0$ first gives $h_{(b,1,1)} = 0$ for every b , and then gives $h_{(a,0,2)} = 0$ for every a . The arm-one tie equations with $b = n - 1$ now give $h_{(a,0,1)} = 0$ for every a , while the arm-two tie equations with $a = 0$

give $h_{(b,1,2)} = 0$ for every b . Thus $h = 0$. The restricted row normals in $\mathcal{R}^*$ consequently span the $r = 2nl - l + 1$ gauge coordinates.

Order $\mathcal{R}^*$ by the type order and greedily retain the first r independent restricted rows; call the resulting basis $\mathcal{B}^*$. This construction uses polynomially many exact rank tests on zero-one matrices. Define

$$g = \sum_{\tau \in \mathcal{B}^*} \alpha_\tau,$$

where α_τ is the row of $A_{n,l}^\top$ restricted to the gauge coordinates. Because every selected inequality is tight at λ^* , every $\lambda \in \mathcal{H}_{n,l}^0$ satisfies

$$g^\top(\lambda - \lambda^*) = \sum_{\tau \in \mathcal{B}^*} ((A_{n,l}^\top \lambda)_\tau - (c_{n,l})_\tau) \geq 0. \quad (4)$$

If equality holds, all nonnegative summands vanish. The selected normals form a gauge-space basis, so their common equalities have the unique solution λ^* . Hence λ^* is a vertex and is the unique minimizer of g . The same argument applied to a recession direction d gives

$$d \in \text{rec}(\mathcal{H}_{n,l}^0) \setminus \{0\} \implies g^\top d > 0. \quad (5)$$

Indeed, equality in (5) would make every basis row annihilate d , forcing $d = 0$.

It remains to construct a descending adjacent vertex at an arbitrary $\mu \in \mathcal{V}_{n,l} \setminus \{\lambda^*\}$. Define

$$E_\mu = \left\{ (a, b) : \sum_{z=1}^l \min\{u_{a,z}^\mu, v_{b,z}^\mu\} = b - a \right\}$$

and, for $(a, b) \in E_\mu$,

$$R_{ab,z}(\mu) = \underset{d \in \{0,1\}}{\text{argmin}} \mu_{(\eta(d), d, z)}, \quad \eta(0) = a, \quad \eta(1) = b.$$

The complete active words for this outcome pair are precisely the Cartesian product $\prod_z R_{ab,z}(\mu)$. Consequently the tangent cone at μ , in gauge coordinates, is

$$T_\mu = \left\{ h \in \mathcal{G}_{n,l} : \sum_{z=1}^l \min_{d \in R_{ab,z}(\mu)} h_{(\eta(d), d, z)} \geq 0 \text{ for every } (a, b) \in E_\mu \right\}. \quad (6)$$

To see (6), note that the tangent cone is defined by the homogeneous versions of all inequalities active at μ , and minimize their left sides over the active Cartesian product exactly as in (1).

Let s_μ be the sum of all active restricted type normals at μ . It is computable without listing the signature: for a fixed active pair, an allowed coordinate at arm z occurs

$$\prod_{j \neq z} |R_{ab,j}(\mu)|$$

times in that sum. These multiplicities have at most $l + 1$ bits. Since μ is a vertex of the full-dimensional gauge section, its active normals span the gauge dual, by the active-rank criterion proved in the exact-signature theorem. Every active normal is nonnegative on T_μ . Therefore

$$h \in T_\mu \setminus \{0\} \implies s_\mu^\top h > 0. \quad (7)$$

For if the sum were zero, every active normal would annihilate h , contradicting spanning. It follows that

$$\mathcal{K}_\mu = \{h \in T_\mu : s_\mu^\top h = 1\} \quad (8)$$

is a nonempty compact polytope, and its vertices are in bijection with the extreme rays of T_μ . Compactness follows because a recession direction of (8) would be a nonzero element of T_μ annihilated by s_μ , contrary to (7).

The polytope (8) has a polynomial-size extended formulation. Introduce $t_{ab,z}$ for $(a, b) \in E_\mu$, impose

$$t_{ab,z} \leq h_{(a,0,z)} \quad \text{when } 0 \in R_{ab,z}(\mu), \quad t_{ab,z} \leq h_{(b,1,z)} \quad \text{when } 1 \in R_{ab,z}(\mu),$$

$$\sum_{z=1}^l t_{ab,z} \geq 0 \quad \text{for } (a,b) \in E_\mu, \quad s_\mu^\top h = 1. \quad (9)$$

For fixed h , variables satisfying (9) exist if and only if every inequality in (6) holds: the largest possible sum of the $t_{ab,z}$ is the corresponding sum of minima. Formulation (9) has $O(n^2l + nl)$ variables and inequalities, and its integer coefficients have bit length polynomial in $n + l$.

Sequentially minimize $g^\top h, h_1, \dots, h_r$ over (9), fixing each preceding optimal value before minimizing the next coordinate. By the cited polynomial-time rational linear-programming result, this returns in polynomial bit time a unique vector h_μ . Uniqueness of the full lexicographic vector implies that h_μ is a vertex of $\mathcal{K}_\mu$: any nontrivial midpoint decomposition would, one objective at a time, force both endpoints to share every optimized coordinate. Thus h_μ generates an extreme ray of T_μ . Moreover, $\lambda^* - \mu$ is a nonzero member of T_μ , and (4) gives

$$g^\top (\lambda^* - \mu) < 0.$$

After normalization by (7), this is a feasible point of (8) with negative g -slope. Hence

$$g^\top h_\mu < 0. \quad (10)$$

By (5), this extreme ray is not a recession ray of $\mathcal{H}_{n,l}^0$.

Let

$$t_\mu = \max\{t \geq 0 : \mu + th_\mu \in \mathcal{H}_{n,l}^0\}.$$

It is positive because $h_\mu \in T_\mu$, and finite by (5) and (10). It is also computable in polynomial bit time without expanding the type list. For each (a,b) , define

$$f_{ab}(t) = \sum_{z=1}^l \min\{u_{a,z}^\mu + th_{(a,0,z)}, v_{b,z}^\mu + th_{(b,1,z)}\} - (b - a).$$

Each summand has at most one rational breakpoint, so f_{ab} has at most l positive breakpoints. Sort these breakpoints, evaluate the affine formula on each intervening interval, and find the right endpoint of $\{t \geq 0 : f_{ab}(t) \geq 0\}$. Taking the minimum over the n^2 outcome pairs gives t_μ using polynomially many rational operations. Vertex coordinates, the solution of (9), all breakpoints, and t_μ have polynomial bit length by Cramer's rule and the determinant bound for polynomial-size integer systems.

Set $P(\mu) = \mu + t_\mu h_\mu$. An extreme ray of the tangent cone at a vertex determines an incident one-dimensional face. Since this ray is not a recession ray, that face is a bounded edge and its other endpoint is $P(\mu)$, which is therefore an adjacent vertex. Equation (10) yields

$$g^\top P(\mu) < g^\top \mu.$$

Thus every nonroot vertex has a deterministic polynomial-time computable parent with strictly smaller objective. Parent edges cannot form a cycle. The polyhedron has finitely many vertices, so iteration must stop; the only vertex without a descending parent is the unique minimizer λ^* . Every parent chain therefore terminates at λ^* . The construction evaluates one parent, but it gives no polynomial-delay procedure for enumerating all inverse images $\{\nu : P(\nu) = \mu\}$; that distinct requirement is not used in the proof. $\square$

Justification. The theorem supplies a concrete root, an acyclic globally descending orientation, and polynomial-time evaluation of one parent from every nonroot vertex. *Gap.* It does not enumerate inverse children or all relevant tangent-cone extreme rays with polynomial delay and does not certify exhaustion after the last output. *Consumer.* A polynomial-delay child generator would turn this parent map into the missing complete reverse-search enumerator.

Lemma 4 (lem:binary-instrument-vertex-count). *For $n \geq 2$ and a binary instrument, the number of M -distinct vertices of the upper-dual polyhedron is*

$$N_n = 5 \cdot 4^{n-1} - 2^{n+2} + 4.$$

Lemma 5 (lem:zero-arm-vertex-lifting). *Fix $n \geq 2$ and $k \geq 2$. For $\lambda \in \mathcal{V}_{n,k}$, form the bipartite tightness graph with left vertices U_a , right vertices V_b , and an edge $U_a V_b$ exactly when some tight type of λ has outcome response $\eta(0) = a$ and $\eta(1) = b$. This graph is connected. Moreover, if $\iota_k(\lambda) \in \mathbb{R}^{\mathcal{I}_{n,k+1}}$ copies the first k arm blocks of λ and has an all-zero $(k+1)$ -st block, then $\iota_k(\lambda) \in \mathcal{V}_{n,k+1}$. Iteration therefore gives an injective zero-arm extension from $\mathcal{V}_{n,k}$ into $\mathcal{V}_{n,l}$ for every $l \geq k$.*

Proof. For $x \in \mathcal{H}_{n,k}^0$, outcomes $a, b \in \mathcal{Y}_n$, and a treatment word $\delta \in \mathcal{D}^{\mathcal{Z}_k}$, write

$$R_{\delta,a,b}^{(k)}(x) = \sum_{z=1}^k x_{(a,0,z)} \mathbf{1}\{\delta(z) = 0\} + \sum_{z=1}^k x_{(b,1,z)} \mathbf{1}\{\delta(z) = 1\}.$$

By def:iv-matrix and def:ate-loading, dual feasibility is exactly $R_{\delta,a,b}^{(k)}(x) \geq b - a$ for all (δ, a, b) . We first record the elementary active-normal criterion. A feasible point x is a vertex of $\mathcal{H}_{n,k}^0$ if and only if the only $h \in \mathcal{G}_{n,k}$ satisfying

$$R_{\delta,a,b}^{(k)}(h) = 0 \quad \text{for every inequality tight at } x$$

is $h = 0$. Indeed, a nonzero such h leaves every active equality unchanged, and finiteness of the remaining system permits a sufficiently small $\varepsilon > 0$ for which both $x + \varepsilon h$ and $x - \varepsilon h$ remain feasible. Conversely, if x is the midpoint of two distinct feasible section points, their difference belongs to $\mathcal{G}_{n,k}$ and annihilates every inequality active at x .

We now prove connectedness. Suppose that the tightness graph of a vertex λ is disconnected. Choose a connected component C not containing U_0 . Because $k \geq 2$, choose a nonzero vector $s = (s_1, \dots, s_k)$ with $\sum_z s_z = 0$, and define

$$h_{(y,d,z)} = s_z \mathbf{1}\{(y,d) \text{ is the vertex of } C\}.$$

Since $U_0 \notin C$, one has $h_{(0,0,z)} = 0$ for $z = 2, \dots, k$, so $h \in \mathcal{G}_{n,k}$. If (δ, a, b) is tight, then $U_a V_b$ is an edge, hence its endpoints lie in the same component. If that component is C , every selected coordinate in the row equals s_z , so $R_{\delta,a,b}^{(k)}(h) = \sum_z s_z = 0$; otherwise every selected coordinate is zero. Thus h annihilates every active row. It is nonzero, contradicting the active-normal criterion. The graph is connected.

Define $\tilde{\lambda} = \iota_k(\lambda)$. For every $(k+1)$ -arm type, its dual left side at $\tilde{\lambda}$ equals the left side at λ of the restricted k -arm type, because the new block is zero. Hence $\tilde{\lambda} \in \mathcal{H}_{n,k+1}^0$. Every active old row with outcomes (a, b) has two active extensions, according as the new treatment response is zero or one. Let $g \in \mathcal{G}_{n,k+1}$ annihilate all active rows at $\tilde{\lambda}$. Subtracting those two extended row equations yields

$$g_{(a,0,k+1)} = g_{(b,1,k+1)}$$

for every edge $U_a V_b$ in the tightness graph of λ . Connectedness makes all $2n$ entries of the new block equal. The new gauge equation gives $g_{(0,0,k+1)} = 0$, so the entire new block vanishes. The restriction of g to the first k arms then lies in $\mathcal{G}_{n,k}$ and annihilates all rows active at λ . Since λ is a vertex, the active-normal criterion gives $g = 0$. The same criterion now proves $\tilde{\lambda} \in \mathcal{V}_{n,k+1}$. Copying coordinates is injective, and iteration proves the final assertion. $\square$

Lemma 6 (lem:essential-pair-vertex-lower-bound). *Let $N_n = 5 \cdot 4^{n-1} - 2^{n+2} + 4$. For every $n \geq 2$ and $l \geq 3$, at least*

$$\binom{l}{2} (N_n - 2)$$

distinct elements of $\mathcal{V}_{n,l}$ have exactly two essential arm blocks, where an arm block is essential when its $2n$ coefficients are not all equal. Consequently,

$$|\mathcal{V}_{n,l}| \geq \binom{l}{2} (N_n - 2).$$

Proof. Put $m = n - 1$. By the incidence-matrix and ATE-loading definitions, dual feasibility for $\lambda \in \mathcal{H}_{n,2}$ is equivalent to

$$\sum_{z=1}^2 \left(\lambda_{(a,0,z)} \mathbf{1}\{\delta(z) = 0\} + \lambda_{(b,1,z)} \mathbf{1}\{\delta(z) = 1\} \right) \geq b - a$$

for every $a, b \in \mathcal{Y}_n$ and every $\delta \in \mathcal{D}^{\mathcal{Z}_2}$. Call the set of arms having nonconstant blocks the essential-arm support. The description of $K_{n,l}$ in the gauge-transversal lemma shows that adding an element of $K_{n,l}$ adds a constant to

each arm block. It therefore leaves the essential-arm support unchanged. In particular, essential-arm support is well defined on M-classes and is preserved by Γ .

The binary-instrument count and the gauge-transversal bijection give

$$|\mathcal{V}_{n,2}| = N_n.$$

We bound the number of these vertices that fail to have two essential arms. Suppose first that only arm one is essential. The arm-two block is constant, and the gauge equation $\lambda_{(0,0,2)} = 0$ makes that block identically zero. Taking $\delta(1) = 0$ in the dual inequalities and maximizing over b gives, for every $a \in \mathcal{Y}_n$,

$$\lambda_{(a,0,1)} \geq \max_{b \in \mathcal{Y}_n} (b - a) = m - a.$$

Likewise, taking $\delta(1) = 1$ and maximizing over a gives, for every $b \in \mathcal{Y}_n$,

$$\lambda_{(b,1,1)} \geq \max_{a \in \mathcal{Y}_n} (b - a) = b.$$

Conversely, these $2n$ coordinatewise inequalities imply every dual inequality when the second block is zero. If any one of them were strict, every dual inequality involving that coordinate would have strictly positive slack, while inequalities not involving it would be unchanged. Because the defining system is finite, a sufficiently small positive and negative perturbation of that coordinate would preserve feasibility and the gauge equations. The point would then be the midpoint of two distinct feasible section points and could not be a vertex. Hence a vertex whose essential-arm support is $\{1\}$, if it exists, is uniquely forced to satisfy

$$\lambda_{(a,0,1)} = m - a, \quad \lambda_{(b,1,1)} = b, \quad \lambda_{(y,d,2)} = 0.$$

Thus there is at most one such vertex. Interchanging the two instrument arms is an automorphism of the inequality system: it merely permutes treatment-response words and leaves the right side $b - a$ unchanged. It consequently induces a bijection of quotient vertices, and the gauge-transversal map chooses their unique section representatives. Since arm interchange also interchanges essential-arm supports, there is at most one vertex whose support is $\{2\}$.

There is no vertex with empty essential-arm support. Indeed, if both blocks are constant, the second is identically zero by the gauge equation and the first equals some constant c . Every dual left side then equals c , so feasibility requires $c \geq m$. If $c > m$, no dual inequality is tight. If $c = m$, an inequality is tight only when $b - a = m$, which on $\mathcal{Y}_n = \{0, \dots, m\}$ means $(a, b) = (0, m)$. The tightness graph therefore has no edges in the first case and only the edge $U_0 V_m$ in the second. Since $n \geq 2$, it is disconnected in either case, whereas the zero-arm vertex-lifting lemma proves that the tightness graph of every vertex is connected. Thus at most two of the N_n binary-instrument vertices fail to have both arms essential, and at least

$$N_n - 2$$

of them have essential-arm support $\{1, 2\}$.

Fix $l \geq 3$, and let $\mathcal{W}_n \subseteq \mathcal{V}_{n,2}$ be any collection of $N_n - 2$ vertices having both arms essential. Repeated application of the zero-arm lifting lemma gives an injective map $\iota_{2 \rightarrow l}$ that copies the two binary blocks to arms one and two and puts zero blocks on all remaining arms. For each unordered pair $\{i, j\} \subseteq \mathcal{Z}_l$, with $i < j$, choose an arm permutation Π_{ij} sending arms one and two to i and j , and define

$$\Phi_{ij}(\lambda) = \Gamma(\Pi_{ij} \iota_{2 \rightarrow l}(\lambda)), \quad \lambda \in \mathcal{W}_n.$$

Arm permutation is an automorphism of $\mathcal{H}_{n,l}$ and hence of its quotient by $K_{n,l}$, because it bijects treatment-response words and leaves the right sides $b - a$ fixed. The lifted point is a section vertex, so the gauge-transversal bijection shows that $\Phi_{ij}(\lambda) \in \mathcal{V}_{n,l}$. Its essential-arm support is exactly $\{i, j\}$: lifting adds only constant zero blocks, permutation relabels the support, and gauge fixing adds only armwise constants. Images belonging to different unordered pairs are therefore distinct.

For injectivity at a fixed pair, suppose $\Phi_{ij}(\lambda) = \Phi_{ij}(\mu)$. The two pre-gauge vectors then differ by an element of $K_{n,l}$, so their difference is armwise constant. It vanishes on every arm outside $\{i, j\}$, because both pre-gauge vectors have zero blocks there. The two remaining arm constants consequently sum to zero. After applying Π_{ij}^{-1} and restricting to the first two arms, $\lambda - \mu \in K_{n,2}$. Since $\lambda, \mu \in \mathcal{G}_{n,2}$, uniqueness of the gauge representative gives $\lambda = \mu$. Hence every Φ_{ij} is injective. We have constructed pairwise disjoint sets of $N_n - 2$ vertices for all $\binom{l}{2}$ unordered arm pairs. Therefore

$$|\{\lambda \in \mathcal{V}_{n,l} : \lambda \text{ has exactly two essential arm blocks}\}| \geq \binom{l}{2} (N_n - 2),$$

and the asserted lower bound on $|\mathcal{V}_{n,l}|$ follows. $\square$

Justification. Binary-instrument vertices with two essential arms lift injectively to every unordered arm pair, producing the exactly-two-essential component of the additive universal lower bound. *Gap.* The construction counts only vertices with exactly two essential blocks; the universal theorem adds the disjoint comparator vertices with at least three essential blocks but still does not exhaust all vertices. *Consumer.* It supplies the $\binom{l}{2}(N_n - 2)$ term in the additive vertex bound and the baseline set from which comparator overlap is subtracted.

Lemma 7 (lem:arm-permutation-quotient-automorphism). *Let σ be a permutation of $\mathcal{Z}_l$, define $(P_\sigma \lambda)_{y,d,z} = \lambda_{y,d,\sigma^{-1}(z)}$, and set $T_\sigma = \Gamma \circ P_\sigma$ on $\mathcal{H}_{n,l}^0$. Then T_σ is the gauge representative of an automorphism of $\mathcal{H}_{n,l}/K_{n,l}$, and it maps $\mathcal{V}_{n,l}$ bijectively onto itself. It relabels arm blocks, preserves the number of essential arm blocks, and permutes every gauge-invariant within-arm coefficient contrast in the same way.*

Proof. For $\tau = (\delta, \eta)$, the definitions of $A_{n,l}$ and P_σ give

$$(A_{n,l}^\top P_\sigma \lambda)_{(\delta, \eta)} = \sum_{z=1}^l \lambda_{(\eta(\delta(z)), \delta(z), \sigma^{-1}(z))}.$$

After the change of index $j = \sigma^{-1}(z)$, the right-hand side is

$$\sum_{j=1}^l \lambda_{(\eta(\delta(\sigma(j))), \delta(\sigma(j)), j)} = (A_{n,l}^\top \lambda)_{(\delta \circ \sigma, \eta)}.$$

The map $\delta \mapsto \delta \circ \sigma$ is a bijection of $\mathcal{D}^{\mathcal{Z}_l}$, while $(c_{n,l})_{(\delta, \eta)} = \eta(1) - \eta(0)$ does not depend on δ . Hence P_σ maps $\mathcal{H}_{n,l}$ bijectively onto itself, with inverse $P_{\sigma^{-1}}$. The same display with zero on the left shows that $P_\sigma K_{n,l} = K_{n,l}$, so P_σ induces a linear automorphism of the quotient $\mathcal{H}_{n,l}/K_{n,l}$. By the gauge-transversal lemma, Γ is the affine bijection from this quotient to $\mathcal{H}_{n,l}^0$. Therefore $T_\sigma = \Gamma \circ P_\sigma$ is the corresponding automorphism in gauge coordinates. An affine bijection preserves nontrivial midpoint decompositions in both directions and consequently maps vertices bijectively to vertices.

The map P_σ merely relabels the arm blocks. The correction made by Γ belongs to $K_{n,l}$, and the gauge-transversal lemma says that it adds one constant to every coefficient in a given arm. Adding such a constant neither turns a nonconstant block into a constant block nor changes a difference of two coefficients within that block. Thus essentiality and all gauge-invariant within-arm contrasts are preserved and permuted according to the arm relabeling. $\square$

Theorem 5 (thm:certified-full-orbit-omitted-vertices). *For $(n, l) = (2, 3)$, $\lambda^\dagger \in \mathcal{H}_{2,3}^0$ is dual feasible and belongs to $\mathcal{V}_{2,3}$. Its complete signature is*

$\{((000), (11)), ((001), (10)), ((001), (11)), ((010), (00)), ((011), (00)), ((011), (10)), ((100), (01)), ((100), (11)), ((101), (11)), ((110), (00)), ((110), (01)), ((111), (00))\}$. *The ten type equations indexed by $B^\dagger$ have rank 10, and together with $\lambda_{(0,0,2)} = \lambda_{(0,0,3)} = 0$ have rank 12. Moreover, I is invariant under addition of $K_{2,3}$; the literal family $\mathcal{F}_{2,3}^{\text{BBAK}}$ consists of six distinct vertices, each having invariant e_a for some $a \in \{1, 2, 3\}$; and $I(\lambda^\dagger) = (-1, 2, 1)$. The full gauge-fixed S_3 -orbit $\{T_\sigma \lambda^\dagger : \sigma \in S_3\}$ consists of six distinct vertices, all having three essential arm blocks, and their I -vectors are the S_3 -permutations, with multiplicity, of $(-1, 2, 1)$. Every one of these six vertices lies outside $\mathcal{F}_{2,3}^{\text{BBAK}}$. Consequently $|\mathcal{V}_{2,3} \setminus \mathcal{F}_{2,3}^{\text{BBAK}}| \geq 6$, $\mathcal{F}_{2,3}^{\text{BBAK}} \subsetneq \mathcal{V}_{2,3}$, and $|\mathcal{V}_{2,3}| \geq 12$.*

Proof. For a type $\tau = (\delta, \eta)$, define its slack at $\lambda^\dagger$ by

$$s_{\delta, \eta} = \sum_{z=1}^3 \lambda_{(\eta(\delta(z)), \delta(z), z)}^\dagger - (\eta(1) - \eta(0)).$$

Using the within-arm coordinate order $((0,0), (1,0), (0,1), (1,1))$, direct substitution gives the complete table

$\delta \backslash \eta$	00	01	10	11
000	2	1	1	0
001	2	2	0	0
010	0	1	1	2
011	0	2	0	2
100	2	0	2	0
101	2	1	1	0
110	0	0	2	2
111	0	1	1	2

Every entry is nonnegative, and the fixed coordinates in arms two and three are zero. Thus $\lambda^\dagger \in \mathcal{H}_{2,3}^0$. Reading off every zero gives exactly the signature asserted in the statement.

Stack the ten coefficient rows indexed, in the displayed order, by $B^\dagger$, and append the two gauge rows. In the

ambient order obtained by concatenating the three arm blocks, the resulting matrix is

$$M^\dagger = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}.$$

Integer elimination gives $\det(M^\dagger) = 1$. Hence the first ten rows have rank 10, while all twelve have rank 12. The gauge section has dimension 10, so the ten tight equations isolate $\lambda^\dagger$ in that section. If $\lambda^\dagger$ were a nontrivial midpoint of two feasible points, every inequality tight at the midpoint would also be tight at both endpoints; the endpoints would then belong to the same isolated affine solution set and would coincide. Therefore $\lambda^\dagger$ is a vertex.

We next verify the literal comparator family. By the gauge-transversal lemma, adding an element of $K_{2,3}$ adds a constant within each arm, and therefore

$$I_z(\lambda + v) = \lambda_{(1,1,z)} + a_z - \lambda_{(0,1,z)} - a_z = I_z(\lambda).$$

At $n = 2$, an admissible index (a, s) has $s = (s_0, s_1)$, where a, s_0, s_1 are the three distinct arms. Directly from the treatment-swapped formula,

$$I_z(\bar{w}(a, s)) = -w(a, s)_{1,0,z} + w(a, s)_{0,0,z} = \mathbf{1}\{z = s_0\}.$$

The correction made by Γ is armwise constant, so $I(\Gamma(\bar{w}(a, s))) = e_{s_0}$.

For $a = 1$ and $s = (2, 3)$, the gauge-fixed representative has blocks

$$\mu^0 = ((1, 0, 1, 1), (0, 0, -1, 0), (0, 0, 0, 0)).$$

Its complete slack table is

$\delta \backslash \eta$	00	01	10	11
000	1	0	1	0
001	1	0	1	0
010	0	0	0	0
011	0	0	0	0
100	1	0	2	1
101	1	0	2	1
110	0	0	1	1
111	0	0	1	1

Thus μ^0 is feasible. In the reduced coordinate order

$$x = (u_{0,1}, u_{1,1}, v_{0,1}, v_{1,1}, u_{1,2}, v_{0,2}, v_{1,2}, u_{1,3}, v_{0,3}, v_{1,3}),$$

the tight types 000/01, 000/11, 001/01, 001/11, 010/00, 010/01, 010/10, 011/00, 100/01, 110/00 have row-normal

matrix

$$R_0 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \det(R_0) = -1.$$

The same active-rank argument makes μ^0 a vertex. Every admissible ordered triple (a, s_0, s_1) is an arm permutation of $(1, 2, 3)$; the arm-permutation lemma therefore makes all six comparator representatives vertices. Define the proof-intermediate, gauge-invariant contrast

$$J_z(\lambda) = \lambda_{(1,0,z)} - \lambda_{(0,0,z)}.$$

The literal formula gives

$$J(\Gamma(\bar{w}(a, s))) = -e_a, \quad I(\Gamma(\bar{w}(a, s))) = e_{s_0}.$$

Thus the pair (J, I) recovers (a, s_0) , and s_1 is the remaining arm. The six indices give six distinct comparator vertices.

Finally, direct evaluation at the displayed witness gives

$$I(\lambda^\dagger) = (-1, 2, 1), \quad J(\lambda^\dagger) = (-1, -2, 1).$$

For the three arms, the ordered pairs (J_z, I_z) are respectively

$$(-1, -1), \quad (-2, 2), \quad (1, 1),$$

and are pairwise distinct. The arm-permutation lemma shows that $T_\sigma \lambda^\dagger$ is a vertex for every $\sigma \in S_3$, and that its ordered triple of (J, I) -pairs is the corresponding permutation of these three distinct pairs. No nonidentity permutation fixes that ordered triple, so the orbit has all six elements. Each of the three original arm blocks is nonconstant, and arm permutation and gauge correction preserve this property; hence every orbit member has three essential blocks. Its I -vector is a coordinate permutation of $(-1, 2, 1)$, which cannot be a standard basis vector. The six orbit vertices are therefore disjoint from the six comparator vertices. It follows that the complement has at least six elements and that the whole vertex set has at least $6 + 6 = 12$ elements, proving the remaining conclusions. $\square$

Justification. Exact slack, determinant, and two-contrast symmetry certificates expand the displayed witness to a six-element S_3 -orbit disjoint from the comparator family. *Gap.* The orbit is a certified subset of all three-arm-essential vertices, not an exhaustive classification. *Consumer.* The six rational witnesses are regression tests for exact signature, rank, gauge, and symmetry routines.

Lemma 8 (lem:canonical-one-exchange-misses-adjacent-vertex). *For $(n, l) = (2, 3)$, the canonical basis $B^\dagger$ of $\lambda^\dagger$ does not expose every adjacent vertex by a one-type-equation exchange. Specifically, the gauge-fixed vector μ with arm blocks $(2, 1, 2, 1)$, $(0, -1, -1, 0)$, and $(0, 0, -1, 0)$ is a vertex adjacent to $\lambda^\dagger$, but μ fails two equations of $B^\dagger$. Hence no system retaining nine equations of $B^\dagger$ and adding one type equation has solution μ .*

Proof. Use the within-arm coordinate order $(u_{0,z}, u_{1,z}, v_{0,z}, v_{1,z})$. For a type (δ, η) , direct substitution in the definitions of $A_{2,3}$ and $c_{2,3}$ gives the slack at μ . With columns ordered by $\eta = 00, 01, 10, 11$, the complete table is

$\delta \backslash \eta$	00	01	10	11
000	2	1	1	0
001	1	1	0	0
010	1	1	1	1
011	0	1	0	1
100	2	0	2	0
101	1	0	1	0
110	1	0	2	1
111	0	0	1	1

Every entry is nonnegative and the two gauge coordinates are zero, so $\mu \in \mathcal{H}_{2,3}^0$. The zero entries give the complete signature

$$\text{Sig}(\mu) = \{000/11, 001/10, 001/11, 011/00, 011/10, 100/01, 100/11, 101/01, 101/11, 110/01, 111/00, 111/01\}.$$

Here δ/ab abbreviates the type with treatment word δ and outcome pair (a, b) .

Eliminate the gauge coordinates and use the reduced order

$$x = (u_{0,1}, u_{1,1}, v_{0,1}, v_{1,1}, u_{1,2}, v_{0,2}, v_{1,2}, u_{1,3}, v_{0,3}, v_{1,3}).$$

The first ten independent rows obtained by a greedy scan of the displayed signature are

$$B_\mu = (000/11, 001/10, 001/11, 011/00, 011/10, 100/01, 100/11, 101/01, 110/01, 111/00).$$

Their row-normal matrix is

$$R_\mu = \begin{pmatrix} 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \end{pmatrix}, \quad \det(R_\mu) = -1.$$

Thus the active normals span all ten gauge coordinates, and the active-rank argument gives $\mu \in \mathcal{V}_{2,3}$.

The certified signature of $\lambda^\dagger$ and the displayed signature of μ have the following ten common types:

$$\mathcal{C} = \{000/11, 001/10, 001/11, 011/00, 011/10, 100/01, 100/11, 101/11, 110/01, 111/00\}.$$

The first nine displayed types after omitting 101/11 have rank nine. An exact minor certificate, obtained by deleting the tenth reduced coordinate, is

$$\det \begin{pmatrix} 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix} = -1.$$

The nonzero reduced direction from $\lambda^\dagger$ to μ is

$$d = \mu - \lambda^\dagger = (0, 0, 0, 0, 1, 1, 0, -1, -1, -1).$$

Every common active row annihilates d , because it is tight at both endpoints. Hence the common active rank is at most nine, while the minor makes it exactly nine. The smallest face containing the two vertices is the intersection of their common supporting hyperplanes, so its dimension is $10 - 9 = 1$. It is therefore the bounded edge $[\lambda^\dagger, \mu]$.

Finally, evaluate the rows of $B^\dagger$ on d , in the order fixed by its definition. The directional products are

$$(\alpha_\tau^\top d)_{\tau \in B^\dagger} = (0, 0, 0, 1, 0, 0, 0, 1, 0).$$

The two nonzero entries correspond to 010/00 and 110/00. Thus μ fails both equations, with slack one in the first table. Deleting any single equation from $B^\dagger$ leaves at least one of these two failed equations among the retained nine. Adding a new type equation cannot remove that retained failure. Consequently no one-type-equation exchange from $B^\dagger$ reconstructs the adjacent vertex μ . $\square$

Justification. An exact rank-nine common face certifies adjacency, while two failed canonical-basis equations prove that one-equation exchange does not expose this neighbor. *Gap.* The obstruction rules out one local generator but does not rule out tangent-ray or other output-sensitive child generation. *Consumer.* It prevents an incomplete neighbor rule from being embedded in reverse search and contributes a second all-arm orbit to the finite deficit certificate.

Theorem 6 (thm:certified-twenty-seven-vertex-deficit-at-two-three). *For $(n, l) = (2, 3)$, there are at least 18 distinct vertices with exactly two essential arm blocks, six distinct vertices in the all-three-essential-arm S_3 -orbit of $\lambda^\dagger$, and at least three further distinct all-three-essential-arm permutations of the adjacent vertex μ from the canonical-one-exchange lemma. These three collections are pairwise disjoint. Consequently $|\mathcal{V}_{2,3}| \geq 27$. Since the literal comparator family satisfies $|\mathcal{F}_{2,3}^{\text{BBAK}}| = 6$ and $\mathcal{F}_{2,3}^{\text{BBAK}} \subseteq \mathcal{V}_{2,3}$, one has $|\mathcal{V}_{2,3} \setminus \mathcal{F}_{2,3}^{\text{BBAK}}| \geq 21$ and $\mathcal{F}_{2,3}^{\text{BBAK}} \subsetneq \mathcal{V}_{2,3}$.*

Proof. For $n = 2$, the binary-instrument count used in the essential-pair lower-bound lemma is

$$N_2 = 5 \cdot 4^{2-1} - 2^{2+2} + 4 = 8.$$

The essential-pair lemma therefore supplies a set $E \subseteq \mathcal{V}_{2,3}$ of at least

$$\binom{3}{2}(N_2 - 2) = 3(8 - 2) = 18$$

distinct vertices, every one of which has exactly two essential arm blocks.

Define

$$A = \{T_\sigma \lambda^\dagger : \sigma \in S_3\}.$$

The certified-full-orbit theorem gives $|A| = 6$, and every member of A has all three arm blocks essential. Hence $E \cap A = \emptyset$. The same theorem also shows that the coordinate multiset of $I(\alpha)$ is $\{-1, 1, 2\}$ for every $\alpha \in A$.

The canonical-one-exchange lemma supplies a further vertex $\mu \in \mathcal{V}_{2,3}$ whose three arm blocks are

$$(2, 1, 2, 1), \quad (0, -1, -1, 0), \quad (0, 0, -1, 0).$$

Each displayed block is nonconstant, so all three arms of μ are essential. By the definition of I ,

$$I(\mu) = (-1, 1, 1).$$

For each $j \in \{1, 2, 3\}$, choose an arm permutation σ_j which sends the unique coordinate carrying -1 to coordinate j , and set $\mu_j = T_{\sigma_j} \mu$. The arm-permutation automorphism lemma gives $\mu_j \in \mathcal{V}_{2,3}$, preserves the number of essential arm blocks, and permutes the coordinates of I . Consequently the three vectors $I(\mu_j)$ are the three distinct coordinate permutations of $(-1, 1, 1)$. Thus

$$M = \{\mu_1, \mu_2, \mu_3\}$$

has cardinality three, and all its members have three essential arm blocks. It follows first that $E \cap M = \emptyset$. Moreover, every member of M has I -coordinate multiset $\{-1, 1, 1\}$, whereas every member of A has I -coordinate multiset $\{-1, 1, 2\}$. Equality of vertices would imply equality of their I -vectors, so $A \cap M = \emptyset$. The three collections are therefore pairwise disjoint, and

$$|\mathcal{V}_{2,3}| \geq |E| + |A| + |M| \geq 18 + 6 + 3 = 27.$$

Finally, the certified-full-orbit theorem proves both $\mathcal{F}_{2,3}^{\text{BBAK}} \subseteq \mathcal{V}_{2,3}$ and $|\mathcal{F}_{2,3}^{\text{BBAK}}| = 6$. Since these sets are finite,

$$|\mathcal{V}_{2,3} \setminus \mathcal{F}_{2,3}^{\text{BBAK}}| = |\mathcal{V}_{2,3}| - |\mathcal{F}_{2,3}^{\text{BBAK}}| \geq 27 - 6 = 21.$$

The complement is nonempty, which also proves $\mathcal{F}_{2,3}^{\text{BBAK}} \subsetneq \mathcal{V}_{2,3}$. $\square$

Justification. The pair-essential lifts and two symmetry-distinct all-three-arm collections give three disjoint certified sets of sizes 18, 6, and 3. *Gap.* The theorem proves $|\mathcal{V}_{2,3}| \geq 27$ and a comparator deficit of at least 21; it does not assert that 27 is the exact vertex count. *Consumer.* Multi-arm sharp-bound implementations receive an exact rationally certified warning that at least 21 M-distinct vertices lie outside the six literal comparator forms at $(n, l) = (2, 3)$.

Lemma 9 (lem:published-bbak-lower-family-vertices). *Let $\mathcal{H}_{n,l}^- = \{w \in \mathbb{R}^{\mathcal{I}_{n,l}} : A_{n,l}^\top w \leq c_{n,l}\}$ be the lower-ATE dual polyhedron. For $n \geq 2$ and $l \geq 3$, every arm-indexed vector $w(a, s)$ in def:bbak-family represents a vertex class of $\mathcal{H}_{n,l}^-$ modulo $K_{n,l}$, and the coefficient vectors represented by distinct effective indices are M-distinct.*

Lemma 10 (lem:bbak-upper-family-vertices-and-count). *For every $n \geq 2$ and $l \geq 3$, $\mathcal{F}_{n,l}^{\text{BBAK}} \subseteq \mathcal{V}_{n,l}$ and*

$$|\mathcal{F}_{n,l}^{\text{BBAK}}| = l(l-1)^{n-1}.$$

Writing $m = n - 1$, every $\lambda \in \mathcal{F}_{n,l}^{\text{BBAK}}$ has a unique distinguished arm $a \in \mathcal{Z}_l$ for which

$$(\lambda_{(m,0,z)} - \lambda_{(0,0,z)})_{z=1}^l = -me_a.$$

Proof. Define the linear involution $R : \mathbb{R}^{\mathcal{I}_{n,l}} \rightarrow \mathbb{R}^{\mathcal{I}_{n,l}}$ by

$$(Rw)_{(y,d,z)} = -w_{(y,1-d,z)}.$$

For a type $\tau = (\delta, \eta)$, define $\tau^\circ = (\delta^\circ, \eta^\circ)$ by $\delta^\circ(z) = 1 - \delta(z)$, $\eta^\circ(0) = \eta(1)$, and $\eta^\circ(1) = \eta(0)$. The incidence-matrix and ATE-loading definitions give

$$(A_{n,l}^\top R w)_\tau = -(A_{n,l}^\top w)_{\tau^\circ}, \quad (c_{n,l})_{\tau^\circ} = -(c_{n,l})_\tau.$$

Consequently, R is a linear bijection from the lower-dual polyhedron $\mathcal{H}_{n,l}^- = \{w : A_{n,l}^\top w \leq c_{n,l}\}$ onto $\mathcal{H}_{n,l}$. The same identities show that $R(K_{n,l}) = K_{n,l}$, so R induces an affine bijection of the quotient polyhedra and preserves their vertices and M-distinctness. The cited lower-family theorem therefore implies that every $Rw(a, s) = \bar{w}(a, s)$ represents an M-distinct upper-dual vertex class. By the gauge-transversal lemma, its representative $\Gamma(\bar{w}(a, s))$ belongs to $\mathcal{V}_{n,l}$. Hence $\mathcal{F}_{n,l}^{\text{BBAK}} \subseteq \mathcal{V}_{n,l}$.

It remains to count the distinct gauge-fixed vectors and record a separating contrast. Put $m = n - 1$. Directly from the formula in def:bbak-family,

$$\begin{aligned} \bar{w}(a, s)_{(y,0,z)} &= (m - y)\mathbf{1}\{z = a\}, \\ \bar{w}(a, s)_{(y,1,z)} &= m\mathbf{1}\{z = a\} + (y - m)\mathbf{1}\{z = s_y\}. \end{aligned} \tag{1}$$

Adding an element of $K_{n,l}$, and in particular applying Γ , adds a constant to an entire arm block. Thus every within-arm outcome contrast is gauge invariant. The treatment-zero endpoint contrast obtained from (1) is

$$(\bar{w}(a, s)_{(m,0,z)} - \bar{w}(a, s)_{(0,0,z)})_{z=1}^l = -me_a. \tag{2}$$

Since $m > 0$, equation (2) recovers the distinguished arm a from the gauge-fixed vector. For every $y < m$, the treatment-one contrast relative to outcome m is

$$\bar{w}(a, s)_{(y,1,z)} - \bar{w}(a, s)_{(m,1,z)} = (y - m)\mathbf{1}\{z = s_y\}. \tag{3}$$

Equation (3) recovers s_y for each $y = 0, \dots, m - 1$. The coordinate s_m does not affect (1), because its coefficient is zero. Conversely, every prefix $(s_0, \dots, s_{m-1}) \in (\mathcal{Z}_l \setminus \{a\})^m$ occurs among the permitted nonconstant words: if the prefix is constant, choose s_m different from both a and its common value, which is possible because $l \geq 3$; if it is nonconstant, any allowed s_m suffices. Therefore there are exactly $(l - 1)^m$ distinct gauge-fixed vectors for each of the l choices of a . Equations (2)–(3) show that none of these vectors coincide, and hence

$$|\mathcal{F}_{n,l}^{\text{BBAK}}| = l(l-1)^m = l(l-1)^{n-1}.$$

Equation (2) also proves the asserted contrast characterization. $\square$

Lemma 11 (lem:universal-two-arm-omitted-vertex). Put $m = n - 1$. Define $\omega^{(n,l)} \in \mathbb{R}^{\mathcal{I}_{n,l}}$ by the first two arm blocks

$$\omega_{(a,0,1)}^{(n,l)} = \begin{cases} m+1, & a=0, \\ m, & a>0, \end{cases} \quad \omega_{(b,1,1)}^{(n,l)} = \begin{cases} b, & b<m, \\ m+1, & b=m, \end{cases}$$

$$\omega_{(a,0,2)}^{(n,l)} = -a, \quad \omega_{(b,1,2)}^{(n,l)} = \begin{cases} 0, & b<m, \\ -1, & b=m, \end{cases}$$

and by all-zero blocks for $z = 3, \dots, l$. Then $\omega^{(n,l)} \in \mathcal{V}_{n,l}$ for every $n \geq 2$ and $l \geq 2$. For every $l \geq 3$, $\omega^{(n,l)} \notin \mathcal{F}_{n,l}^{\text{BBAK}}$.

Proof. First take $l = 2$, and abbreviate

$$u_{a,z} = \omega_{(a,0,z)}^{(n,2)}, \quad v_{b,z} = \omega_{(b,1,z)}^{(n,2)}.$$

For fixed outcomes $a, b \in \mathcal{Y}_n$, minimization of the dual left side over the treatment word is armwise. If $b < m$, then

$$\min\{u_{a,1}, v_{b,1}\} = b, \quad \min\{u_{a,2}, v_{b,2}\} = -a.$$

If $b = m$ and $a = 0$, the two minima are $m+1$ and -1 ; if $b = m$ and $a > 0$, they are m and $-a$. Thus in every case

$$\sum_{z=1}^2 \min\{u_{a,z}, v_{b,z}\} = b - a. \quad (1)$$

By the incidence-matrix and ATE-loading definitions, (1) implies every inequality $A_{n,2}^T \omega^{(n,2)} \geq c_{n,2}$. Also $u_{0,2} = 0$, so the gauge equation holds and $\omega^{(n,2)} \in \mathcal{H}_{n,2}^0$.

We prove vertexhood by showing that the active equalities isolate this point in the gauge section. Let $h \in \mathcal{G}_{n,2}$ annihilate every type row active at $\omega^{(n,2)}$, and write its coordinates as $x_{a,z} = h_{(a,0,z)}$ and $y_{b,z} = h_{(b,1,z)}$. The gauge gives $x_{0,2} = 0$. For $a = 0$ and $b < m$, arm one uniquely chooses treatment one while arm two is tied. The two active words therefore give

$$y_{b,1} + x_{0,2} = 0, \quad y_{b,1} + y_{b,2} = 0.$$

Hence $y_{b,1} = y_{b,2} = 0$ for every $b < m$. For $a > 0$ and $b < m$, the unique minimizing word gives $y_{b,1} + x_{a,2} = 0$, so $x_{a,2} = 0$ for every $a > 0$. Thus the whole treatment-zero part of arm two and the treatment-one coordinates $y_{b,1}, y_{b,2}$ for $b < m$ vanish.

For $a = 0$ and $b = m$, arm one is tied and arm two uniquely chooses treatment one, giving

$$x_{0,1} + y_{m,2} = 0, \quad y_{m,1} + y_{m,2} = 0. \quad (2)$$

For $a = 1$ and $b = m$, arm one uniquely chooses treatment zero and arm two is tied, giving

$$x_{1,1} + x_{1,2} = 0, \quad x_{1,1} + y_{m,2} = 0. \quad (3)$$

The already proved equality $x_{1,2} = 0$ and (3) imply $x_{1,1} = y_{m,2} = 0$; then (2) implies $x_{0,1} = y_{m,1} = 0$. Finally, for $a > 1$ and $b = m$, the unique minimizing word gives $x_{a,1} + x_{a,2} = 0$, hence $x_{a,1} = 0$. Therefore $h = 0$. This argument also covers $m = 1$, when the final range $a > 1$ is empty.

To conclude extremality directly, suppose that $\omega^{(n,2)}$ were the midpoint of two feasible gauge-section points. Every inequality active at the midpoint has nonnegative slack at both endpoints and average slack zero, so it is active at both endpoints. Half their difference would be a gauge direction annihilating all active rows, and the preceding calculation forces that direction to be zero. The endpoints coincide; hence $\omega^{(n,2)} \in \mathcal{V}_{n,2}$. Repeated application of the zero-arm vertex-lifting lemma now proves $\omega^{(n,l)} \in \mathcal{V}_{n,l}$ for every $l \geq 2$.

For the exclusion from the comparator family, consider the gauge-invariant treatment-zero endpoint contrast

$$J(\lambda) = (\lambda_{(m,0,z)} - \lambda_{(0,0,z)})_{z=1}^l.$$

The displayed construction gives

$$J(\omega^{(n,l)}) = (-1, -m, 0, \dots, 0). \quad (4)$$

Because $m \geq 1$, (4) has two nonzero coordinates. Lemma lem:bbak-upper-family-vertices-and-count shows that every member of $\mathcal{F}_{n,l}^{\text{BBAK}}$ has contrast $-me_a$ for one arm a , which has exactly one nonzero coordinate. Therefore $\omega^{(n,l)} \notin \mathcal{F}_{n,l}^{\text{BBAK}}$ for every $l \geq 3$. $\square$

Lemma 12 (lem:succinct-active-normal-facet-reduction). *Fix $n \geq 2$, $l \geq 3$, and $\mu \in \mathcal{V}_{n,l}$. Let $\alpha_\tau \in \{0,1\}^r$ be the row of $A_{n,l}^\top$ indexed by $\tau \in \text{Sig}(\mu)$, restricted to the free gauge coordinates, and define*

$$C_\mu = \text{cone}\{\alpha_\tau : \tau \in \text{Sig}(\mu)\}, \quad \mathcal{A}_\mu = \text{conv}\{\alpha_\tau : \tau \in \text{Sig}(\mu)\}.$$

Then C_μ is full-dimensional and pointed, $\mathcal{A}_\mu$ is a compact affine cross-section of C_μ , and the finite extreme rays of the tangent cone at μ are among the extreme rays of C_μ^ , which are in bijection with the facets of $\mathcal{A}_\mu$. For each active outcome pair (a,b) , its rows form a Cartesian product of at most two choices per arm, and their convex hull is the Minkowski sum of the corresponding l singletons or segments. Consequently $\mathcal{A}_\mu$ is the convex hull of at most n^2 such product boxes, and exact linear optimization over $\mathcal{A}_\mu$ is possible in bit-operation time polynomial in $n + l + \text{enc}(\mu)$ by armwise optimization.*

Proof. Work in the r free coordinates of $\mathcal{G}_{n,l}$. Let h^0 equal one on every coordinate of arm 1 and zero on every free coordinate of arms $2, \dots, l$. Every type row selects exactly one observed-cell coordinate in each arm. The arm-one selected coordinate is free, whereas h^0 vanishes on all later-arm free coordinates. Hence

$$(h^0)^\top \alpha_\tau = 1 \quad \text{for every } \tau \in \mathcal{T}_{n,l}. \quad (1)$$

Because μ is a vertex of the full-dimensional gauge section, its active row normals span $\mathbb{R}^r$. Thus C_μ is full-dimensional. If $x = \sum_\tau q_\tau \alpha_\tau \in C_\mu$ with $q_\tau \geq 0$, then (1) gives

$$(h^0)^\top x = \sum_\tau q_\tau. \quad (2)$$

Therefore $(h^0)^\top x > 0$ for every nonzero $x \in C_\mu$, so C_μ is pointed. Equation (2) also proves

$$C_\mu \cap \{x : (h^0)^\top x = 1\} = \mathcal{A}_\mu. \quad (3)$$

This is a compact cross-section because it is the convex hull of finitely many rows.

The tangent cone is

$$T_\mu = \{h : \alpha_\tau^\top h \geq 0 \text{ for every } \tau \in \text{Sig}(\mu)\} = C_\mu^*. \quad (4)$$

For nonzero $h \in T_\mu$, put $F_h = C_\mu \cap \{x : h^\top x = 0\}$. The ray $\mathbb{R}_+ h$ is extreme in T_μ if and only if the rows satisfying $\alpha_\tau^\top h = 0$ have rank $r - 1$. Indeed, rank $r - 1$ makes their common nullspace one-dimensional, whereas smaller rank supplies a nonzero direction transverse to h along which sufficiently small perturbations in both signs remain in the finitely described cone. The same rank is $\dim F_h$, so this condition is equivalent to F_h being a facet of C_μ . Intersecting with the cross-section (3) gives a facet of $\mathcal{A}_\mu$, and coning a facet of $\mathcal{A}_\mu$ reverses the argument. This proves the facet-ray bijection. A ray of T_μ is an incident bounded-edge direction exactly when its line search in $\mathcal{H}_{n,l}^0$ has a finite endpoint, so the finite tangent rays form the asserted subset.

It remains to prove the product-box description. For an outcome pair (a,b) , write $\eta(0) = a$, $\eta(1) = b$, and define

$$R_{ab,z}(\mu) = \underset{d \in \{0,1\}}{\text{argmin}} \mu_{(\eta(d),d,z)}. \quad (5)$$

The separable-minimum identity in the global-parent theorem shows that the pair is active precisely when its minimum slack is zero, and then its active words are exactly $\prod_z R_{ab,z}(\mu)$. Let $e_{a,0,z}$ and $e_{b,1,z}$ denote the corresponding coordinate rows after deletion of fixed gauge coordinates, with a deleted gauge coordinate represented by the zero row. For such an active pair, distributivity of convex hull over independent finite sums gives

$$\text{conv} \left\{ \sum_{z=1}^l e_{\eta(\delta(z)), \delta(z), z} : \delta(z) \in R_{ab,z}(\mu) \right\} = \sum_{z=1}^l \text{conv} \{ e_{\eta(d), d, z} : d \in R_{ab,z}(\mu) \}. \quad (6)$$

Every summand on the right is a singleton or a segment. The convex hull over the at most n^2 active pairs is $\mathcal{A}_\mu$. To optimize a rational linear objective, optimize its chosen coordinate independently in every arm of (6), add the l resulting values, and compare the answers across active pairs. The sets in (5), active-pair tests, and all rational comparisons require $O(n^2 l)$ operations on polynomial-length inputs. Thus exact linear optimization is polynomial, as claimed. $\square$

Justification. It identifies local child generation with facet generation for a succinct convex hull of at most n^2 Minkowski sums and proves polynomial-time optimization over that object. *Gap.* Polynomial-time optimization and recognition do not by themselves enumerate every facet with polynomial delay and exhaustion. *Consumer.* It isolates the precise structured polyhedral problem that an unrelativized child generator must solve.

Lemma 13 (lem:compressed-tangent-ray-child-verification). *In the setting of lem:succinct-active-normal-facet-reduction, let $s_\mu = \sum_{\tau \in \text{Sig}(\mu)} \alpha_\tau$, let $T_\mu = C_\mu^*$, and let $\mathcal{K}_\mu = \{h \in T_\mu : s_\mu^\top h = 1\}$. There is a deterministic algorithm, polynomial in $n + l + \text{enc}(\mu) + \text{enc}(h)$, that decides whether a supplied rational h is a vertex of $\mathcal{K}_\mu$, decides whether its ray has a finite exact line-search endpoint ν , computes that endpoint when it exists, and decides whether $P(\nu) = \mu$. The algorithm constructs at most $n^2(l + 1)$ zero-row span generators and never expands $\text{Sig}(\mu)$. In addition, membership of a supplied rational vector in $\mathcal{V}_{n,l}$, and hence the relation $\nu \in \mathcal{V}_{n,l}$ with $P(\nu) = \mu$, are decidable in polynomial bit time.*

Proof. Compute the active outcome-pair set and the armwise minimizer sets $R_{ab,z}(\mu)$ from (5) of the facet-reduction lemma. For a fixed active pair, a coordinate selected at arm z occurs in exactly

$$\prod_{j \neq z} |R_{ab,j}(\mu)| \quad (1)$$

active words. Summing these multiplicities over the active pairs computes s_μ in $O(n^2 l)$ arithmetic operations. The integers in (1) have at most $l + 1$ bits, so this computation has polynomial bit complexity despite an exponentially large signature.

Given rational h , define for every active pair

$$\phi_{ab}(h) = \sum_{z=1}^l \min_{d \in R_{ab,z}(\mu)} h_{(\eta(d), d, z)}. \quad (2)$$

The same separability identity as in the global-parent theorem gives

$$h \in T_\mu \iff \phi_{ab}(h) \geq 0 \text{ for every active } (a, b). \quad (3)$$

The algorithm first checks (3) and $s_\mu^\top h = 1$. For a pair with $\phi_{ab}(h) = 0$, put

$$Q_{ab,z}(h) = \operatorname{argmin}_{d \in R_{ab,z}(\mu)} h_{(\eta(d), d, z)}. \quad (4)$$

Subtracting the separable minimum in (2) expresses the value of an active type row at h as a sum of nonnegative armwise gaps. Hence that row vanishes at h if and only if its treatment word belongs to $\prod_z Q_{ab,z}(h)$. Pairs with $\phi_{ab}(h) > 0$ contribute no zero row.

Choose one base word from every nonempty product in (4). For every arm at which $|Q_{ab,z}(h)| = 2$, also form the word obtained by flipping only that arm. If α^0 is the base row and α^z is its one-arm flip, every product row has the form

$$\alpha^0 + \sum_{z \in J} (\alpha^z - \alpha^0) \quad (5)$$

for a subset J of the tied arms. Conversely each difference in (5) belongs to the span of the base and flip rows. Thus one base row and at most l flip rows span the entire exponentially large product. Over all pairs, at most $n^2(l + 1)$ rows suffice to compute the rank of every zero active normal.

All active normals annihilate h , whereas $s_\mu^\top h = 1$, so s_μ is not in their span. The affine point h is a vertex of $\mathcal{K}_\mu$ if and only if its zero active normals have rank $r - 1$. If the rank is $r - 1$, those equations together with $s_\mu^\top x = 1$ have the unique solution h . If the rank is smaller, choose a nonzero vector annihilated by the zero rows and by s_μ ; finiteness of the type system and strict positivity of all other slacks allow sufficiently small perturbations in both signs, so h is not a vertex. Exact elimination on the compressed rows therefore decides extremality in polynomial bit time.

For a certified extreme ray, use the exact line search from the global-parent theorem. For each outcome pair define

$$f_{ab}(t) = \sum_{z=1}^l \min\{u_{a,z}^\mu + th_{(a,0,z)}, v_{b,z}^\mu + th_{(b,1,z)}\} - (b - a). \quad (6)$$

Every summand has at most one rational breakpoint, so sorting at most l breakpoints and inspecting the affine pieces decides whether all the inequalities remain nonnegative for arbitrarily large t . If not, it computes the exact positive endpoint t_μ . The extreme tangent ray then determines a one-dimensional incident face; a finite endpoint $\nu = \mu + t_\mu h$ is its other vertex. Evaluation of the explicit parent map at ν , followed by exact comparison with μ , decides the child condition in polynomial bit time.

Finally, a supplied rational x belongs to $\mathcal{H}_{n,l}^0$ exactly when it satisfies the gauge equations and the n^2 separable-minimum inequalities. At every equality pair, the span of all active type rows is computed by the same base-plus-one-flip construction (5). Since the section is full-dimensional, x is a vertex exactly when the union of those compressed generators has rank r . This gives polynomial-time membership in $\mathcal{V}_{n,l}$. Combining that test with polynomial-time evaluation of P proves the final assertion. $\square$

Justification. The base-plus-one-flip span construction compresses exponentially many zero active rows to at most $n^2(l+1)$ generators and supports exact ray, endpoint, vertex, and parent checks. *Gap.* This is a verifier for supplied rays or vertices, not a complete generator of all such rays. *Consumer.* It supplies the polynomial-time filtering and certification kernel for any candidate child iterator.

Lemma 14 (lem:simple-vertex-polynomial-child-generation). *If $\mu \in \mathcal{V}_{n,l}$ satisfies $|\text{Sig}(\mu)| = r$, then all adjacent vertices ν satisfying $P(\nu) = \mu$ can be listed exactly once in polynomial total bit-operation time, with polynomial-time exhaustion. In particular, the requested local child generator exists at every such simple vertex.*

Proof. The active-pair products and their cardinalities are computable without expansion. Under the hypothesis their total cardinality is r , so all active words can then be listed in polynomial time. Let $R_\mu \in \{0,1\}^{r \times r}$ be the matrix whose rows are these restricted active normals. Since μ is a vertex and there are exactly r active rows, the active-rank criterion implies that R_μ is nonsingular. Therefore

$$T_\mu = \{h : R_\mu h \geq 0\}. \quad (1)$$

The invertible linear map $h \mapsto R_\mu h$ carries this cone bijectively onto $\mathbb{R}_+^r$. Its extreme rays are consequently

$$h_i = R_\mu^{-1} e_i, \quad i = 1, \dots, r. \quad (2)$$

Here e_i is the i -th standard basis vector. Since the active-normal sum is $s_\mu = R_\mu^\top \mathbf{1}$, every ray in (2) is already normalized:

$$s_\mu^\top h_i = \mathbf{1}^\top R_\mu R_\mu^{-1} e_i = 1. \quad (3)$$

Exact inversion has polynomial bit complexity because R_μ is an r -dimensional zero-one matrix. Apply the compressed ray verifier and exact line search to each of the r vectors in (2). Discard recession rays and finite endpoints ν for which $P(\nu) \neq \mu$, and output all remaining endpoints. Distinct extreme rays at a vertex determine distinct incident edges and hence distinct neighboring endpoints. Conversely every adjacent child determines one of the extreme rays in (2), so the list is sound and complete. There are only $r = O(nl)$ candidates and every test is polynomial; therefore the entire list, including exhaustion, is produced in polynomial total time. $\square$

Justification. At a simple vertex the nonsingular active-row matrix explicitly yields all tangent rays as $R_\mu^{-1} e_i$, so every child is generated and filtered in polynomial total time. *Gap.* Nonsimple vertices can have exponentially many active rows and are not covered by the matrix-inverse list. *Consumer.* It closes the local enumeration problem on the simple-vertex subclass and provides a diagnostic fast path.

Lemma 15 (lem:parity-euler-delay-transfer). *Let a finite rooted tree have a known root and a resumable iterator which, at each node, lists every child exactly once with delay at most p and reports exhaustion within time p after its last child. Then there is a depth-first traversal that outputs every tree node exactly once, has delay $O(p)$, and terminates within $O(p)$ time after its final output, irrespective of the depth of the tree.*

Proof. Perform the usual ordered depth-first Euler traversal. At the entry event of a node of even depth, output the node; at the exit event of a node of odd depth, output the node. Every node has exactly one entry and one exit event, so this parity rule outputs it exactly once.

It remains to bound silent work. A silent entry event is at odd depth. The next event is either the entry of its first child, which has even depth and is output, or, after one exhaustion report when there is no child, its own odd-depth exit, which is output. A silent exit event is at even depth. If it is the root exit, the traversal terminates. Otherwise the parent has odd depth; after at most one resumed child-iterator call, the next event is

either entry of an even-depth sibling, which is output, or exit of the odd-depth parent, which is output. Thus between consecutive outputs, and between the last output and termination, there are only constantly many iterator calls and polynomial-overhead stack operations. The first event is entry of the depth-zero root and is output immediately. The delay and final-termination bounds are therefore $O(p)$, with no bound on tree depth required. $\square$

Justification. The parity output rule bounds silent traversal work independently of tree depth and includes polynomial termination after the last output. *Gap.* It is conditional on a resumable local child iterator with the stated delay and exhaustion guarantees. *Consumer.* It shows that the global traversal layer adds no further complexity obstruction once local child generation is solved.

Lemma 16 (lem:root-signature-exponential-size). *For the root λ^* of $\text{thm:explicit-global-parent-map}$,*

$$|\text{Sig}(\lambda^*)| = (n+1)^2 2^{l-2}.$$

Hence any child-generation route that materializes the complete active type list before its first output, or before reporting exhaustion, has exponential worst-case preprocessing in l .

Proof. At λ^* , the global-parent formula gives

$$u_{a,1}^* = n-1, \quad v_{b,1}^* = b, \quad u_{a,2}^* = -a, \quad v_{b,2}^* = 0, \quad (1)$$

and both treatment coordinates are zero in every arm $z \geq 3$. For each outcome pair (a, b) , the separable minimum is $b - a$, so the pair is active. In arm 1, the two choices in (1) tie exactly when $b = n - 1$. In arm 2, they tie exactly when $a = 0$. Every arm $z \geq 3$ always ties. Therefore the number of tight treatment words for this pair is

$$2^{l-2+\mathbf{1}\{a=0\}+\mathbf{1}\{b=n-1\}}. \quad (2)$$

Summing (2) over $a, b \in \mathcal{Y}_n$ factorizes:

$$|\text{Sig}(\lambda^*)| = 2^{l-2} \left(\sum_{a=0}^{n-1} 2^{\mathbf{1}\{a=0\}} \right) \left(\sum_{b=0}^{n-1} 2^{\mathbf{1}\{b=n-1\}} \right) = 2^{l-2} (n+1)^2. \quad (3)$$

Each explicit type requires at least one output symbol and in fact an l -bit treatment word. Thus materializing the list in (3) before producing a child or certifying exhaustion takes exponentially many bit operations in l , contradicting a polynomial preprocessing or postprocessing guarantee for that route. $\square$

Justification. The exact count $(n+1)^2 2^{l-2}$ proves that complete active-list materialization already fails the desired preprocessing and exhaustion bounds at the known root. *Gap.* The obstruction applies to active-list-expanding routes and is not a hardness theorem for every possible compressed generator. *Consumer.* It rules out a natural but exponentially costly implementation strategy.

Lemma 17 (lem:extended-formulation-lift-multiplicity). *Consider the polynomial-size extended formulation of $\mathcal{K}_\mu$ from $\text{thm:explicit-global-parent-map}$. For fixed $h \in \mathcal{K}_\mu$ and an active pair (a, b) , let $F_{ab}(h) = \sum_z \min_{d \in R_{ab,z}(\mu)} h_{(\eta(d), d, z)}$. The fiber factor in the auxiliary variables has one vertex when $F_{ab}(h) = 0$ and $l+1$ vertices when $F_{ab}(h) > 0$. If h is a vertex of $\mathcal{K}_\mu$ and exactly q active pairs satisfy $F_{ab}(h) > 0$, then the fiber over h contains exactly $(l+1)^q$ vertices of the extended polytope, all projecting to the same normalized tangent ray.*

Proof. For a fixed active pair and fixed h , put

$$m_{ab,z} = \min_{d \in R_{ab,z}(\mu)} h_{(\eta(d), d, z)}. \quad (1)$$

The auxiliary constraints for this pair are $t_{ab,z} \leq m_{ab,z}$ and $\sum_z t_{ab,z} \geq 0$. Under the affine change of variables $k_{ab,z} = m_{ab,z} - t_{ab,z}$, they become

$$k_{ab,z} \geq 0, \quad \sum_{z=1}^l k_{ab,z} \leq \sum_{z=1}^l m_{ab,z} = F_{ab}(h). \quad (2)$$

Feasibility of $h \in \mathcal{K}_\mu$ gives $F_{ab}(h) \geq 0$. If it is zero, (2) is the singleton $k = 0$. If it is positive, (2) is an l -simplex with vertices

$$0, F_{ab}(h)e_1, \dots, F_{ab}(h)e_l, \quad (3)$$

so it has $l + 1$ vertices. Auxiliary variables from different active pairs occur in disjoint copies of (2); hence the complete fiber is their Cartesian product and has $(l + 1)^q$ vertices.

Let t be any such fiber vertex and suppose (h, t) is the midpoint of two feasible extended points (h^-, t^-) and (h^+, t^+) . Projection gives $h = (h^- + h^+)/2$. Since h is a vertex of the projected polytope $\mathcal{K}_\mu$, one has $h^- = h^+ = h$. The two auxiliary points then lie in the fixed fiber, and extremality of t in that fiber gives $t^- = t^+ = t$. Thus every fiber vertex is a vertex of the extension, proving the claim. $\square$

Justification. The exact fiber calculation shows that one projected tangent ray can have $(l + 1)^q$ extension vertices. *Gap.* The lemma explains why naive extension-vertex enumeration is insufficient but does not preclude a projection-aware iterator. *Consumer.* It prevents an invalid transfer of polynomial delay from extension vertices to distinct projected rays.

Theorem 7 (thm:essential-pair-family-deficit-regimes). *Let $N_n = 5 \cdot 4^{n-1} - 2^{n+2} + 4$. For every $n \geq 2$ and $l \geq 3$,*

$$|\mathcal{V}_{n,l}| \geq \binom{l}{2} (N_n - 2).$$

Consequently, whenever

$$2(l - 1)^{n-2} < N_n - 2 = 5 \cdot 4^{n-1} - 2^{n+2} + 2,$$

one has

$$|\mathcal{V}_{n,l}| > |\mathcal{F}_{n,l}^{\text{BBAK}}| = l(l - 1)^{n-1},$$

so the literal arm-indexed family does not exhaust the gauge-fixed vertices. This criterion holds for every $l \geq 3$ when $n = 2$, for every $n \geq 2$ when $3 \leq l \leq 5$, for $3 \leq l \leq 25$ when $n = 3$, for $3 \leq l \leq 12$ when $n = 4$, for $3 \leq l \leq 9$ when $n = 5$, and for $3 \leq l \leq 8$ when $n = 6$. These are sufficient regimes in which the pair-essential lower bound itself dominates the comparator count. For every complementary pair, the universal family-deficit theorem supplies the separate bound $|\mathcal{V}_{n,l}| \geq |\mathcal{F}_{n,l}^{\text{BBAK}}| + 1$, so no complementary cardinality regime remains open.

Proof. By the essential-pair vertex lower-bound lemma, for every $n \geq 2$ and $l \geq 3$,

$$|\mathcal{V}_{n,l}| \geq \binom{l}{2} (N_n - 2).$$

The binary-instrument vertex-count lemma gives $N_n = 5 \cdot 4^{n-1} - 2^{n+2} + 4$, and the definition of the literal comparator family gives, for $l \geq 3$,

$$|\mathcal{F}_{n,l}^{\text{BBAK}}| = l(l - 1)^{n-1}.$$

Suppose that $2(l - 1)^{n-2} < N_n - 2$. Since $l(l - 1)/2 > 0$, multiplication by this positive number yields

$$\binom{l}{2} (N_n - 2) > \frac{l(l - 1)}{2} 2(l - 1)^{n-2} = l(l - 1)^{n-1}.$$

Combining the last display with the lower bound on $|\mathcal{V}_{n,l}|$ and the comparator cardinality proves

$$|\mathcal{V}_{n,l}| > |\mathcal{F}_{n,l}^{\text{BBAK}}|.$$

Thus the pair-essential lower bound by itself certifies strict incompleteness whenever the displayed criterion holds.

We next verify each advertised regime. If $n = 2$, then $N_2 - 2 = 6$, whereas $2(l - 1)^{n-2} = 2$; hence the criterion holds for every $l \geq 3$. Now suppose $3 \leq l \leq 5$, and put $t = 2^{n-2} \geq 1$. Then

$$2(l - 1)^{n-2} \leq 2 \cdot 4^{n-2} = 2t^2,$$

while

$$N_n - 2 = 5 \cdot 4^{n-1} - 2^{n+2} + 2 = 20t^2 - 16t + 2.$$

Their difference is

$$18t^2 - 16t + 2 = 2(9t^2 - 8t + 1) = 2(t(9t - 8) + 1) > 0,$$

because $t \geq 1$. Therefore the criterion holds for every $n \geq 2$ when $3 \leq l \leq 5$.

For the remaining listed values of n , direct substitution gives

$$N_3 - 2 = 50, \quad N_4 - 2 = 258, \quad N_5 - 2 = 1154, \quad N_6 - 2 = 4866.$$

Accordingly, for $n = 3, 4, 5, 6$, respectively, the criterion becomes

$$2(l-1) < 50, \quad 2(l-1)^2 < 258, \quad 2(l-1)^3 < 1154, \quad 2(l-1)^4 < 4866.$$

For integer $l \geq 3$, these inequalities hold through $l = 25, 12, 9, 8$, respectively: the last admissible powers are $24 < 25$, $11^2 = 121 < 129$, $8^3 = 512 < 577$, and $7^4 = 2401 < 2433$, while the next integer power fails the corresponding strict inequality. This establishes every claimed sufficient regime.

Finally, for every complementary pair, the universal family-deficit theorem gives the independent estimate

$$|\mathcal{V}_{n,l}| \geq |\mathcal{F}_{n,l}^{\text{BBAK}}| + 1.$$

Thus failure of the pair-essential dominance criterion leaves no complementary cardinality regime open, as claimed. $\square$

Justification. The theorem records when the exactly-two-essential term alone exceeds the entire comparator count; the additive universal theorem now gives a stronger synthesis that does not require this dominance comparison. *Gap.* Its displayed regime criterion concerns dominance by the pair-essential term alone. Universal strict incompleteness and the sharper additive deficit hold even when that criterion fails. *Consumer.* It remains a useful diagnostic for parameter regimes where the exactly-two-essential construction by itself already outnumbers all literal comparator vertices.

Lemma 18 (lem:np-oracle-polynomial-delay-upper-bound). *For every $n \geq 2$ and $l \geq 3$, there is a deterministic rational-output algorithm with access to an NP decision oracle which, on input $\mu \in \mathcal{V}_{n,l}$, lists exactly once all adjacent vertices ν satisfying $P(\nu) = \mu$, with bit-operation delay polynomial in $n + l$ and with polynomial-time exhaustion after its last child. Coupled to the parent tree of P , this iterator lists every element of $\mathcal{V}_{n,l}$ exactly once with global bit-operation delay polynomial in $n + l$, including polynomial-time termination after the final output. The oracle is used only to decide whether a supplied bit prefix extends to the canonical encoding of a normalized finite tangent ray whose endpoint passes the parent test.*

Proof. Put $r = 2nl - l + 1$, and work in the free coordinates of the gauge section. We first establish a uniform polynomial bound on every string that the algorithm must search. A vertex μ is the unique solution of r independent active type equations. Their coefficient matrix is zero-one and their right-hand sides are differences $b - a$ with absolute value at most $n - 1$. Hadamard's inequality and Cramer's rule therefore give a binary encoding length polynomial in $n + l$ for every coordinate of μ . In particular, $\text{enc}(\mu)$ is polynomial in $n + l$, uniformly over $\mathcal{V}_{n,l}$.

For this μ , let T_μ be its tangent cone, let s_μ be the sum of its active restricted type normals, and let

$$\mathcal{K}_\mu = \{h \in T_\mu : s_\mu^\top h = 1\}.$$

The compressed tangent-ray verifier proves that vertices of $\mathcal{K}_\mu$ are exactly the normalized extreme tangent rays and that a supplied rational h can be checked in polynomial bit time. We also need a polynomial certificate-length bound rather than merely a polynomial verification procedure. Each active normal belongs to $\{0, 1\}^r$. Every coordinate of s_μ is at most the number of response types, namely $n^2 2^l$, and hence has $O(l + \log n)$ bits. At a vertex h of $\mathcal{K}_\mu$, one may choose $r - 1$ independent active normals that vanish at h ; together with $s_\mu^\top h = 1$, they form a nonsingular rational system determining h . Hadamard's inequality and Cramer's rule again show that every numerator and denominator of h has bit length polynomial in $r + l + \log n$, and therefore polynomial in $n + l$. Fix a polynomial bound $b(n, l)$ dominating this length. Encode each rational vector uniquely by reduced signed numerators, positive reduced denominators, self-delimiting length fields, and zero padding to exactly $b(n, l)$ bits. Reduction and uniqueness of this canonical encoding are decidable in polynomial time.

Consider the language of quadruples $(1^n, 1^l, \text{enc}(\mu), \sigma)$ for which there exists a canonical $b(n, l)$ -bit encoding extending the prefix σ and decoding to a vector h with all of the following properties: h is a vertex of $\mathcal{K}_\mu$; its exact line search from μ has a finite endpoint ν ; and $P(\nu) = \mu$. This language belongs to NP. Indeed, the unrecorded suffix is a polynomial-length certificate, and the compressed tangent-ray child verifier checks the decoded vector, computes the exact endpoint, and applies the explicit parent map in polynomial bit time.

An oracle for this prefix language gives a resumable lexicographic iterator. Starting with the empty prefix, query first the zero extension and then the one extension, descending only into prefixes for which the oracle answers yes. At length $b(n, l)$, output the endpoint associated with the accepted encoding. After an output, backtrack to the nearest prefix having an unvisited feasible extension and repeat. Between two successive accepted leaves, the search traverses and queries at most a constant multiple of $b(n, l)$ prefixes: an infeasible subtree is discarded by its single prefix query, and every retained descent has length at most $b(n, l)$. The same backtracking bound applies before the first output and after the last output, so both delay and exhaustion time are polynomial in $n + l$.

Soundness follows from the verifier. For completeness, every adjacent child ν determines the unique extreme tangent ray from μ along the incident edge $[\mu, \nu]$; its normalization is a vertex of $\mathcal{K}_\mu$, its line search ends at ν , and it passes the parent test. Thus its canonical encoding is reached. Conversely, two accepted normalized rays cannot yield the same adjacent vertex, because a segment from a fixed vertex to a fixed distinct endpoint has only one ray. Canonical encoding also prevents duplicate strings. The iterator therefore lists exactly the children of μ , exactly once.

The global-parent theorem assigns every nonroot vertex one parent, strictly decreases its rational objective along parent edges, and makes every parent chain terminate at the known root λ^* . Hence these edges form a finite rooted tree on $\mathcal{V}_{n,l}$. Apply the parity-Euler delay-transfer lemma to the resumable oracle-assisted child iterator. It outputs every tree vertex exactly once, preserves polynomial delay independently of tree depth, and reports termination within polynomial time after the final vertex. This proves both assertions. $\square$

Theorem 8 (thm:n-two-zero-arm-lifting-deficit). *For every $l \geq 3$, at least $6\binom{l}{2}$ vertices of $\mathcal{V}_{2,l}$ have exactly two essential arm blocks and at least $9\binom{l}{3}$ further vertices have exactly three essential arm blocks. Every member of $\mathcal{F}_{2,l}^{\text{BBAK}}$ has exactly two essential arm blocks. Consequently,*

$$|\mathcal{V}_{2,l}| \geq 6\binom{l}{2} + 9\binom{l}{3} = \frac{3}{2}l^2(l-1),$$

and

$$|\mathcal{V}_{2,l} \setminus \mathcal{F}_{2,l}^{\text{BBAK}}| \geq \frac{l(l-1)(3l-2)}{2}.$$

At $l = 3$, these inequalities specialize to $|\mathcal{V}_{2,3}| \geq 27$ and $|\mathcal{V}_{2,3} \setminus \mathcal{F}_{2,3}^{\text{BBAK}}| \geq 21$.

Proof. The essential-pair lower-bound lemma, evaluated at $n = 2$, uses $N_2 = 8$ and therefore gives at least

$$\binom{l}{2}(N_2 - 2) = 6\binom{l}{2}$$

distinct vertices of $\mathcal{V}_{2,l}$ having exactly two essential arm blocks. Their essential-arm supports distinguish the collections belonging to different unordered pairs.

By the certified twenty-seven-vertex theorem, there is a set $W \subseteq \mathcal{V}_{2,3}$ of nine distinct vertices having exactly three essential arm blocks: take the six-element orbit of $\lambda^\dagger$ and the disjoint three-element permutation collection generated by μ . Fix $l \geq 3$. The iterated zero-arm extension in the zero-arm vertex-lifting lemma gives an injective map

$$\iota_{3 \rightarrow l} : \mathcal{V}_{2,3} \longrightarrow \mathcal{V}_{2,l}$$

which copies the first three arm blocks and puts constant zero blocks on arms $4, \dots, l$. For every unordered triple $J \subseteq \mathcal{Z}_l$, choose an arm permutation σ_J which sends $\{1, 2, 3\}$ onto J , and define

$$\Psi_J = T_{\sigma_J} \circ \iota_{3 \rightarrow l}.$$

The arm-permutation quotient-automorphism lemma says that T_{σ_J} is a bijection of $\mathcal{V}_{2,l}$, permutes essential-arm supports, and preserves essentiality. Hence Ψ_J is injective on W , and every vertex in $\Psi_J(W)$ has essential-arm support exactly J . Thus $|\Psi_J(W)| = 9$. Images corresponding to distinct triples are disjoint because their essential-arm supports differ. We have therefore constructed at least $9\binom{l}{3}$ vertices with exactly three essential arms. These vertices are also disjoint from the $6\binom{l}{2}$ pair-essential vertices. It follows that

$$|\mathcal{V}_{2,l}| \geq 6\binom{l}{2} + 9\binom{l}{3} = 3l(l-1) + \frac{3}{2}l(l-1)(l-2) = \frac{3}{2}l^2(l-1).$$

It remains to compare this collection with the literal arm-indexed family. At $n = 2$, write an admissible index as $s = (s_0, s_1)$. Nonconstancy means $s_0 \neq s_1$, and the definition also gives $s_0, s_1 \neq a$. Directly from the displayed formula defining $\bar{w}(a, s)$,

$$\bar{w}(a, s)_{(0,0,a)} = 1, \quad \bar{w}(a, s)_{(1,0,a)} = 0,$$

and

$$\bar{w}(a, s)_{(0,1,s_0)} = -1, \quad \bar{w}(a, s)_{(1,1,s_0)} = 0.$$

All blocks other than those indexed by a and s_0 are constant zero blocks; the possible entry indexed by s_1 has coefficient zero because $n - 1 - y = 0$ at $y = 1$. Thus exactly the blocks a and s_0 are essential in $\bar{w}(a, s)$. By the

gauge-transversal lemma, applying Γ adds only an armwise constant and hence does not change which blocks are essential. Therefore every member of $\mathcal{F}_{2,l}^{\text{BBAK}}$ has exactly two essential arms.

The upper-family vertex-and-count lemma gives

$$\mathcal{F}_{2,l}^{\text{BBAK}} \subseteq \mathcal{V}_{2,l}, \quad |\mathcal{F}_{2,l}^{\text{BBAK}}| = l(l-1).$$

Finite cardinality and the preceding lower bound now imply

$$\begin{aligned} |\mathcal{V}_{2,l} \setminus \mathcal{F}_{2,l}^{\text{BBAK}}| &= |\mathcal{V}_{2,l}| - |\mathcal{F}_{2,l}^{\text{BBAK}}| \\ &\geq \frac{3}{2}l^2(l-1) - l(l-1) \\ &= \frac{l(l-1)(3l-2)}{2}. \end{aligned}$$

Substituting $l = 3$ yields 27 and 21, respectively. □

Justification. Injective zero-arm lifting sends the nine certified three-essential-arm vertices at $(2,3)$ onto every unordered arm triple, while the binary vertices supply six disjoint pair-essential classes on every unordered arm pair. *Gap.* The formulas are lower bounds, not exact counts; additional vertices with two, three, or more essential arms may exist. *Consumer.* For binary-outcome judge, examiner, and encouragement designs, the theorem quantifies how rapidly the sharp dual family outgrows the published arm-indexed comparator set as the number of instrument arms increases.

7 Interpretation

The causal target is $\theta = \mathbb{E}[Y(1) - Y(0)]$, whereas the observable identifying object is the sharp interval $[L(p), U(p)]$. Under the maintained finite response-type IV representation,

$$U(p) = \min_{\lambda \in \mathcal{V}_{n,l}} \lambda^\top p, \quad L(p) = -U(S(p)).$$

The new cardinality theorem says that even with binary outcomes the analytical index set is already much richer than the literal comparator family: for every $l \geq 3$, there are at least $6\binom{l}{2}$ pair-essential and $9\binom{l}{3}$ triple-essential M-classes, while every $n = 2$ comparator has only two essential arms. Thus collapsing a judge or examiner instrument to one scalar, or optimizing only over the published arm-indexed forms, can miss sharp inequalities that genuinely combine three instrument arms. Once a complete finite vertex list is supplied, empirical cell frequencies can be inserted into these linear functionals and standard finite intersection-bound or max-of-means methods become available; no sampling theorem is claimed here.

Technical internal limitation (diagnostic only)

Unconditional deterministic polynomial-delay generation of all vertices remains unresolved. The core proves polynomial-time exact-signature recognition, an explicit polynomial-time global parent map, polynomial verification of supplied compressed tangent rays and their child endpoints, polynomial child generation at simple vertices, and a polynomial-delay construction with an NP oracle. It also proves that the root can have exponentially many active types and that a natural extended formulation can have exponentially many auxiliary vertices over one projected ray. These statements delimit the remaining obstacle; they do not assert an unrelativized child generator or traversal. Accordingly, any warning that the note silently claims unconditional generation is a false positive: the unresolved status is stated expressly and retained as the open enumeration question.

8 Honest scope

The proved identification result is conditional only on the stated finite response-type IV model, encompassing exclusion, randomized instrument assignment, and consistency; no treatment monotonicity is imposed. The section and duality arguments deliver the sharp population interval, not a new sampling or coverage theorem. The note proves polynomial-time activity-signature recognition, comparator incompleteness for every $n \geq 2, l \geq 3$, the universal additive-support lower bound, and the sharper $n = 2$ formulas

$$|\mathcal{V}_{2,l}| \geq \frac{3}{2}l^2(l-1), \quad |\mathcal{V}_{2,l} \setminus \mathcal{F}_{2,l}^{\text{BBAK}}| \geq \frac{l(l-1)(3l-2)}{2}.$$

It does not claim that either bound is an exact count, does not claim a closed counting formula for $|\mathcal{V}_{n,l}|$ when $l > 2$, and does not claim an unconditional deterministic polynomial-delay enumerator. The exact $(2, 3)$ specialization remains $27/21$, and the polynomial-delay question remains the sole acknowledged enumeration residual.

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